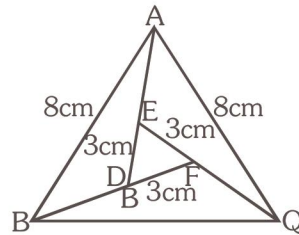
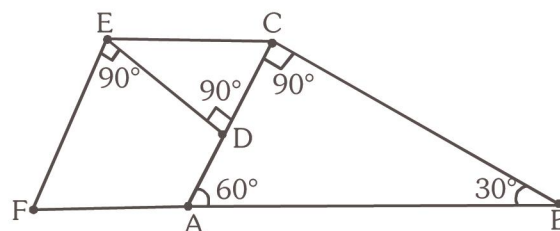


EXERCISE - 4.1

1. ABC is an isosceles triangle with  $AB = AC = 2008$  cm. ADC is drawn as an equilateral triangle on AC outside  $\triangle ABC$ . AD is parallel to BC. The bisector of D meets AB in E, say. Then BE is equal to  
(1) 1004 cm (2) 2008 cm (3) 0 (4) 502 cm
2. In the adjoining figure ABC, DEF are equilateral triangles  $AB = 8$  cm and  $DE = 3$  cm. Then the possible value of  $AE + BD + CF$  is

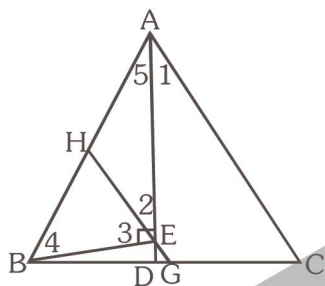


- (1) 6.9 cm (2) 7.1 cm (3) 5.2 cm (4) 8.3 cm
3. The measure of one of the angles of a right triangle is five times that of a second angle. Then the possibility of the second largest angle is  
(1)  $72^\circ$  (2)  $75^\circ$  (3)  $72^\circ$  or  $75^\circ$  (4) None of these
4. Two of the altitudes of the scalene triangle ABC have lengths 4 and 12. If the length of the third altitude is also an integer, then its greatest value can be .....  
(1) 5 (2) 10 (3) 15 (4) 20
5. The perimeter of an isosceles right angled triangle is 2012. Its area is  
(1)  $2012(3 - \sqrt{2})$  (2)  $(1006)^2(3 - \sqrt{2})$  (3)  $(2012)^2$  (4)  $(1006)^2$
6. Triangles ABC and ABD are isosceles with  $AB = AC = BD$ , and AC intersects BD at E. If AC is perpendicular to BD, then  $\angle C + \angle D$  is  
(1)  $115^\circ$  (2)  $120^\circ$  (3)  $130^\circ$  (4)  $135^\circ$
7. In  $\triangle ABC$ ,  $AB = 8$ ,  $BC = 7$ ,  $CA = 6$ . Extend BC to P such that  $\triangle PAB \sim \triangle PCA$ , then the length of PC is  
(1) 7, (2) 8, (3) 9, (4) 10, (E) 11.
8. In the isosceles right triangle ABC,  $\angle B = 90^\circ$ , AD is the median on BC. Write  $AB = BC = a$ . If  $BE \perp AD$ , intersecting AC at E, and  $EF \perp BC$  at F, then EF is  
(1)  $\frac{1}{3}a$ , (2)  $\frac{1}{2}a$ , (3)  $\frac{2}{3}a$ , (4)  $\frac{2}{5}a$ .
9. In the adjoining figure BAC is a  $30^\circ - 60^\circ - 90^\circ$  triangle with  $AB = 20$ . D is the midpoint of AC. The perpendicular at D to AC meets the line parallel to AB through C at E. The line through E perpendicular to DE meets BA produced at F. If  $DF = 5\sqrt{x}$ , then find x.



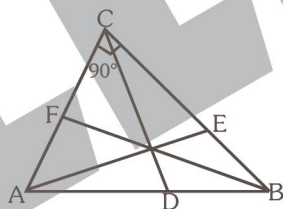
- (1) 1 (2) 7 (3) 5 (4) 2

10. A point P is chosen in the interior of  $\triangle ABC$  such that when lines are drawn through P parallel to the sides of  $\triangle ABC$ , the resulting smaller triangles  $t_1$ ,  $t_2$ , and  $t_3$  in the figure, have areas 4, 9 and 49 respectively. Find the area of  $\triangle ABC$ .
- (1) 144 (2) 140 (3) 130 (4) 131
11. In  $\triangle ABC$ , D, E, F are on the sides BC, CA, AB respectively, such that they are concurrent at a point G,  $BD = 2CD$ , the areas  $S_1 = [GEC] = 3$ ,  $S_2 = [GCD] = 4$ . Find the area of  $\triangle ABC$ .
- (1) 30 (2) 20 (3) 25 (4) 35
12. In  $\triangle ABC$ , the bisector of  $\angle A$  intersects  $\overline{BC}$  at D. A perpendicular to  $\overline{AD}$  from B intersects  $\overline{AD}$  at E. A line segment through E and parallel to  $\overline{AC}$  intersects  $\overline{BC}$  at G, and  $\overline{AB}$  at H. If  $AB = 26$ ,  $BC = 28$ ,  $AC = 30$ , find the measure of  $\overline{DG}$ .



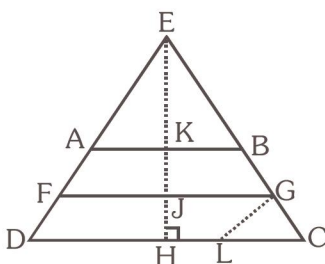
- (1) 2 (2) 3 (3) 1 (4) 4

13. If  $\overline{AE}$  and  $\overline{BF}$  are medians drawn to the legs of right  $\triangle ABC$ , find the numerical value of  $\frac{(AE)^2 + (BF)^2}{(AB)^2}$



- (1)  $\frac{5}{4}$  (2)  $\frac{6}{4}$  (3)  $\frac{7}{4}$  (4)  $\frac{8}{4}$

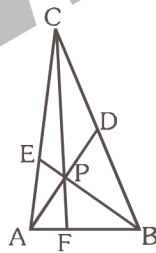
14. The measures of the bases of trapezoid ABCD are 15 and 9, and the measure of the altitude is 4. Legs  $\overline{DA}$  and  $\overline{CB}$  are extended to meet at E. If F is the midpoint of  $\overline{AD}$ , and G is the midpoint of  $\overline{BC}$ , find the area of  $\triangle FGE$ . (The figure is not drawn to scale.)



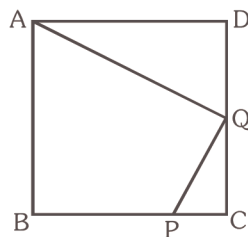
- (1) 48 (2) 49 (3) 46 (4) 52

15. In a right-angled  $\triangle ABC$ ,  $\angle ACB = 90^\circ$ , E, F are on AB such that  $AE = AC$ ,  $BF = BC$ , find  $\angle ECF$  in degrees.

16. Given that the lengths of three sides,  $a, b, c$  of a triangle are positive integers, where  $a < b < c$ . Find the number of the triangles with  $b = 2$ .
17. In a right-angled triangle, if the length of a leg is 21, and the lengths of the other two sides are also positive integers, find the minimum value of its possible perimeter.
18. In right-angled  $\triangle ABC$ ,  $\angle C = 90^\circ$ ,  $E$  is on  $BC$  such that  $AC = BE$ .  $D$  is on  $AB$  such that  $DE \perp BC$ . Given that  $DE + BC = 1$ ,  $BD = \frac{1}{2}$ , find  $\angle B$  in degrees.
19. In  $\triangle ABC$ ,  $AC = BC$ ,  $\angle C = 20^\circ$ ,  $M$  is on the side  $AC$  and  $N$  is on the side  $BC$ , such that  $\angle BAN = 50^\circ$ ,  $\angle ABM = 60^\circ$ . Find  $\angle NMB$  in degrees.
20. In the  $\triangle ABC$ ,  $BE$  is the angle bisector of the  $\angle ABC$ ,  $AD$  is the median on the side  $BC$ , and  $AD$  intersects  $BE$  at  $O$  perpendicularly. Given  $BE = AD = 4$ , find the lengths of three sides of  $\triangle ABC$ .
21. In  $\triangle ABC$ ,  $D$  is the midpoint of  $BC$ ,  $E$  is on  $AC$  such that  $AC = 3EC$ .  $BE$  and  $AD$  intersect at  $G$ . Find  $AG : GD$ .
22. When extending the sides  $AB, BC, CA$  of  $\triangle ABC$  to  $B', C', A'$  respectively, such that  $AB' = 2AB$ ,  $CC' = 2BC$ ,  $AA' = 3CA$ . If area of  $\triangle ABC$  is 1, find the area of  $\triangle A'B'C'$ .
23. Given that the point  $P$  is outside the equilateral triangle  $ABC$  but inside the region of  $\angle ABC$ . If the distances from  $P$  to  $BC, CA, AB$  are  $h_1, h_2$  and  $h_3$  respectively, and  $h_1 - h_2 + h_3 = 6$ , find the area of  $\triangle ABC$ .
24. Given that  $G$  is the centroid of  $\triangle ABC$ ,  $GA = 2\sqrt{3}$ ,  $GB = 2\sqrt{2}$ ,  $GC = 2$ . Find the area of  $\triangle ABC$ .
25. Point  $P$  is inside  $\triangle ABC$ . Line segments  $APD, BPE$ , and  $CPF$  are drawn with  $D$  on  $BC$ ,  $E$  on  $CA$ , and  $F$  on  $AB$  (see the figure below). Given that  $AP = 6$ ,  $BP = 9$ ,  $PD = 6$ ,  $PE = 3$ , and  $CF = 20$ , find the area of  $\triangle ABC$ .

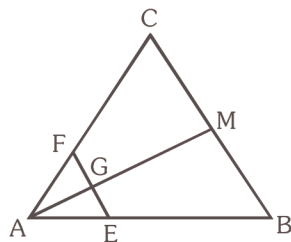


26. In a square  $ABCD$ , let  $P$  be a point on the side  $BC$  such that  $BP = 3PC$  and  $Q$  be the mid-point of  $CD$ . If the area of the triangle  $PCQ$  is 5, what is the area of triangle  $QDA$ ?



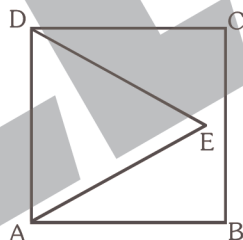
27. A triangle  $\triangle ABC$  is inscribed in a circle of radius 4 cm. Suppose that  $\angle A = 60^\circ$ ,  $AC - AB = 4$  cm, and the area of  $\triangle ABC$  is  $x \text{ cm}^2$ . Find the value of  $\left(\frac{x}{2}\right)^2$ .

28. In  $\triangle ABC$ ,  $AB : AC = 4 : 3$  and  $M$  is the midpoint of  $BC$ .  $E$  is a point on  $AB$  and  $F$  is a point on  $AC$  such that  $AE : AF = 2 : 1$ . It is also given that  $EF$  and  $AM$  intersect at  $G$  with  $GF = 36$  cm and  $GE = x$  cm. Find the value of  $x$ .

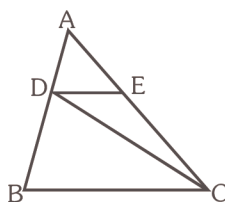


29. Nondegenerate  $\triangle ABC$  has integer side lengths,  $BD$  is an angle bisector,  $AD = 3$ , and  $DC = 8$ . What is the smallest possible value of the perimeter?
30. It is known that the 3 sides of a triangle are consecutive positive integers and the largest angle is twice the smallest angle. Find the perimeter of this triangle.
31. A triangle  $\triangle ABC$  is inscribed in a circle of radius 1, with  $\angle BAC = 60^\circ$ . Altitudes  $AD$  and  $BE$  of  $\triangle ABC$  intersect at  $H$ . Find the smallest possible value of the length of the segment  $AH$ .
32. In triangle  $ABC$ ,  $AB = 28$ ,  $BC = 21$  and  $CA = 14$ . Points  $D$  and  $E$  are on  $AB$  with  $AD = 7$  and  $\angle ACD = \angle BCE$ . Find the length of  $BE$ .
33. Four points in the order  $A, B, C, D$  lie on a circle with the extension of  $AB$  meeting the extension of  $DC$  at  $E$  and the extension of  $AD$  meeting the extension of  $BC$  at  $F$ . Let  $EP$  and  $FQ$  be tangents to this circle with points of tangency  $P$  and  $Q$  respectively. Suppose  $EP = 60$  and  $FQ = 63$ . Determine the length of  $EF$ .
34. The diagram shows an equilateral triangle  $ADE$  inside a square  $ABCD$ . What is the value of

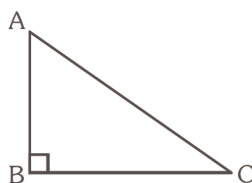
$$\left( \frac{\text{area of } \triangle ADE}{\text{area of } \triangle DEC} \right)^2$$



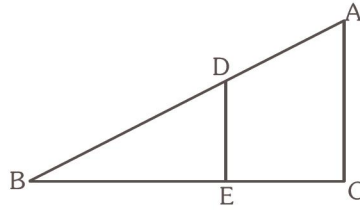
35. In this question,  $S_{\triangle XYZ}$  denotes the area of  $\triangle XYZ$ . In the following figure, if  $DE \parallel BC$ ,  $S_{\triangle ADE} = 1$  and  $S_{\triangle ADC} = 4$ , find  $S_{\triangle DBC}$ .



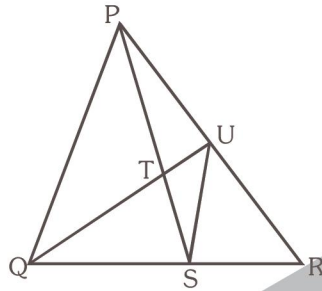
36.  $\triangle ABC$  is a right-angled triangle with  $\angle ABC = 90^\circ$ . A circle  $C_1$  is drawn with  $AB$  as diameter, and another circle  $C_2$  is drawn with  $BC$  as diameter. The circles  $C_1$  and  $C_2$  meet at the points  $B$  and  $P$ . If  $AB = 5$  cm,  $BC = 12$  cm and  $BP = x$  cm, find the value of  $\frac{120}{x}$ .



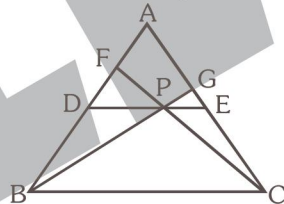
37. In the following diagram,  $\angle ACB = 90^\circ$ ,  $DE \perp BC$ ,  $BE = AC$ ,  $BD = \frac{1}{2}$  cm, and  $DE + BC = 1$  cm. Suppose  $\angle ABC = x^\circ$ . Find the value of  $x$ .



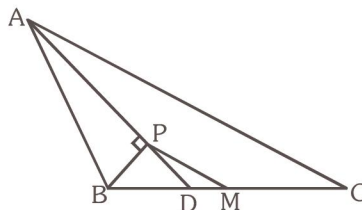
38. In the figure below, S is a point on OR and U is a point on PR. The line segments PS and QU intersect at the point T. It is given that  $PT = TS$  and  $QS = 2RS$ . If the area of  $\triangle PQR$  is  $150 \text{ cm}^2$  and the area of  $\triangle PSU$  is  $x \text{ cm}^2$ . Find the value of  $x$ .



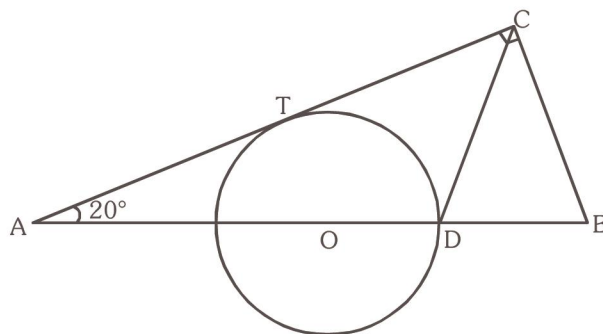
39. A triangle  $\triangle ABC$  has its vertices lying on a circle  $C$  of radius 1, with  $\angle BAC = 60^\circ$ . A circle with center  $I$  is inscribed in  $\triangle ABC$ . The line  $AI$  meets circle  $C$  again at  $D$ . Find the length of the segment  $ID$ .
40. In the triangle  $ABC$ ,  $AB = AC = 1$ ,  $D$  and  $E$  are the midpoints of  $AB$  and  $AC$  respectively. Let  $P$  be a point on  $DE$  and let the extensions of  $BP$  and  $CP$  meet the sides  $AC$  and  $AB$  at  $G$  and  $F$  respectively. Find the value of  $\frac{1}{BF} + \frac{1}{CG}$ .



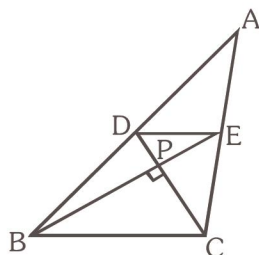
41. In the triangle  $ABC$ ,  $AB = 14$ ,  $BC = 16$ ,  $AC = 26$ ,  $M$  is the midpoint of  $BC$  and  $D$  is the point on  $BC$  such that  $AD$  bisects  $\angle BAC$ . Let  $P$  be the foot of the perpendicular from  $B$  onto  $AD$ . Determine the length of  $PM$ .



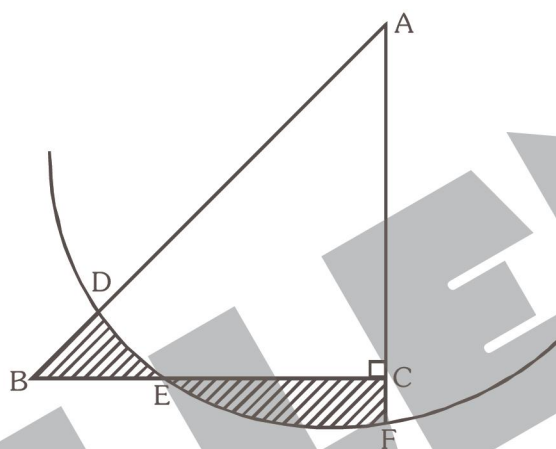
42. In the triangle  $ABC$ ,  $\angle A = 20^\circ$ ,  $\angle C = 90^\circ$ ,  $O$  is a point on  $AB$  and  $D$  is the midpoint of  $OB$ . Suppose the circle centered at  $O$  with radius  $OD$  touches the side  $AC$  at  $T$ . Determine the size of  $\angle BCD$  in degrees.



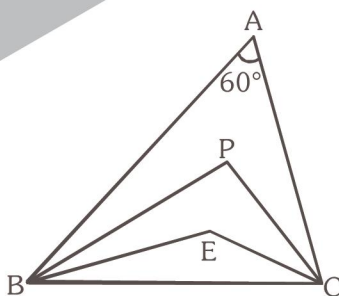
43. In  $\triangle ABC$ , D and E are the midpoints of the sides AB and AC respectively, CD and BE intersect at P with  $\angle BPC = 90^\circ$ . Suppose  $BD = 8\sqrt{5}$  and  $CE = 6\sqrt{5}$ . Find BC.



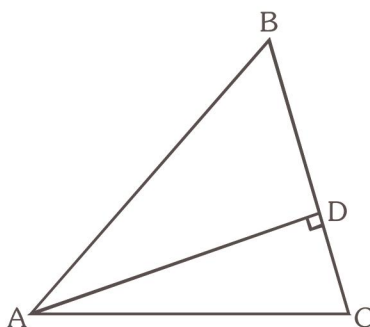
44. In the following right-angled triangle ABC,  $AC = BC = 1$  and DEF is an arc of a circle with center A. Suppose the shaded areas BDE and CEF are equal and  $AD = \frac{x}{\sqrt{\pi}}$ . What is the value of x?



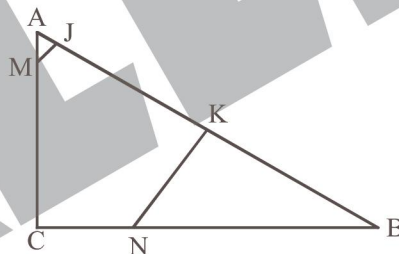
45. The diagram below shows a triangle ABC in which  $\angle A = 60^\circ$ , BP and BE trisect  $\angle ABC$  and CP and CE trisect  $\angle ACB$ . Let the angle  $\angle BPE$  be  $x^\circ$ . Find x.



46. In  $\triangle ABC$   $\angle BAC = 45^\circ$ , D is a point on BC such that AD is perpendicular to BC. If  $BD = 3$  cm and  $DC = 2$  cm, and the area of the  $\triangle ABC$  is  $x$  cm<sup>2</sup>. Find the value of x.

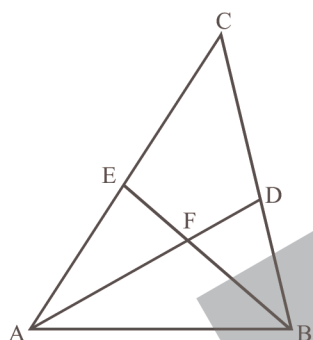


47. In  $\triangle ABC$  (see below),  $AB = AC = \sqrt{3}$  and  $D$  is a point on  $BC$  such that  $AD = 1$ . Find the value of  $BD \cdot DC$ .
48. In  $\triangle ABC$ ,  $\angle CAB = 30^\circ$  and  $\angle ABC = 80^\circ$ . The point  $M$  lies inside the triangle such that  $\angle MAC = 10^\circ$  and  $\angle MCA = 30^\circ$ . Find the value of  $(180^\circ - \angle BMC)$  in degrees.
49. Let  $ABC$  be any triangle. Let  $D$  and  $E$  be the points respectively in the segments of  $AB$  and  $BC$  such that  $AD = 7DB$  and  $BE = 10EC$ . Assume that  $AE$  and  $CD$  meet at point  $F$ . Determine  $[k]$ , where  $k$  is the real number such that  $AF = k \times FE$ .
50. In an acute-angled triangle  $ABC$ , points  $D$ ,  $E$  and  $F$  are the feet of the perpendiculars from  $A$ ,  $B$ , and  $C$  onto  $BC$ ,  $AC$  and  $AB$  respectively. Suppose  $\sin A = \frac{3}{5}$  and  $BC = 39$ , find the length of  $AH$ , where  $H$  is the intersection  $AD$  with  $BE$ .
51. Let  $O$  be the centre of the circumcircle of  $\triangle ABC$ ,  $P$  and  $Q$  the midpoints of  $AO$  and  $BC$ , respectively. Suppose  $\angle CBA = 4\angle OPQ$  and  $\angle ACB = 6\angle OPQ$ . Find the size of  $\angle OPQ$  in degrees.
52. In  $\triangle ABC$ ,  $AC > AB$ , the internal angle bisector of  $\angle A$  meets  $BC$  at  $D$ , and  $E$  is the foot of the perpendicular from  $B$  onto  $AD$ . Suppose  $AB = 5$ ,  $BE = 4$  and  $AE = 3$ . Find the value of the expression  $\left(\frac{AC + AB}{AC - AB}\right) ED$ .
53. Let  $O$  be an interior point of  $\triangle ABC$ . Extend  $AO$  to meet the side  $BC$  at  $D$ . Similarly, extend  $BO$  and  $CO$  to meet  $CA$  and  $AB$  respectively at  $E$  and  $F$ . If  $AO = 30$ ,  $FO = 20$ ,  $BO = 60$ ,  $DO = 10$  and  $CO = 20$ , find  $EO$ .
54. In  $\triangle ABC$ ,  $AB = 13$ ,  $AC = 5$ , and  $BC = 12$ . Points  $M$  and  $N$  lie on  $\overline{AC}$  and  $\overline{BC}$ , respectively, with  $CM = CN = 4$ . Points  $J$  and  $K$  are on  $\overline{AB}$  so that  $\overline{MJ}$  and  $\overline{NK}$  are perpendicular to  $\overline{AB}$ . What is the whole part of area of pentagon  $CMJKN$ .

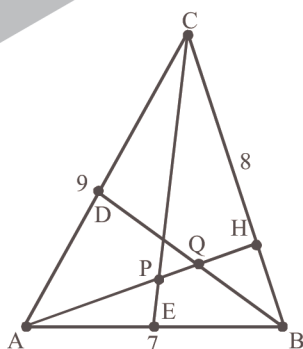


55. In triangle  $ABC$ , side  $AC$  and the perpendicular bisector of  $BC$  meet in point  $D$ , and  $BD$  bisects  $\angle ABC$ . If  $AD = 9$  and  $DC = 7$ , the area of triangle  $ABD = a\sqrt{5}$  then the value of  $a$
56. In  $\triangle ABC$ , we have  $AB = 1$  and  $AC = 2$ . Side  $BC$  and the median from  $A$  to  $BC$  have the same length. If the length of  $BC$  is  $\sqrt{a}$  then the value of  $a$ ?
57. Triangle  $ABC$  with  $AB = 50$  and  $AC = 10$  has area 120. Let  $D$  be the midpoint of  $\overline{AB}$ , and let  $E$  be the midpoint of  $\overline{AC}$ . The angle bisector of  $\angle BAC$  intersects  $\overline{DE}$  and  $\overline{BC}$  at  $F$  and  $G$ , respectively. What is the area of quadrilateral  $FDBG$ ?
58. Triangle  $ABC$  is an isosceles right triangle with  $AB = AC = 3$ . Let  $M$  be the midpoint of hypotenuse  $\overline{BC}$ . Points  $I$  and  $E$  lie on sides  $\overline{AC}$  and  $\overline{AB}$ , respectively, so that  $AI > AE$  and  $AIME$  is a cyclic quadrilateral. Given that triangle  $EMI$  has area 2, the length  $CI$  can be written as  $\frac{a - \sqrt{b}}{c}$ , where  $a$ ,  $b$  and  $c$  are positive integers and  $b$  is not divisible by the square of any prime. What is the value of  $a + b + c$ ?

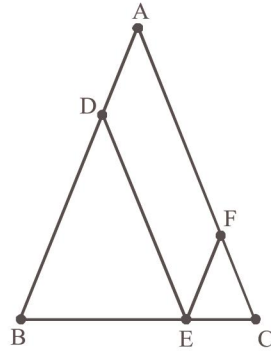
59. In  $\triangle PAT$ ,  $\angle P = 36^\circ$ ,  $\angle A = 56^\circ$ , and  $PA = 10$ . Points  $U$  and  $G$  lie on sides  $\overline{TP}$  and  $\overline{TA}$ , respectively, so that  $PU = AG = 1$ . Let  $M$  and  $N$  be the midpoints of segments  $\overline{PA}$  and  $\overline{UG}$ , respectively. What is the degree measure of the acute angle formed by lines  $MN$  and  $PA$ ?
60. Let  $ABC$  be an equilateral triangle. Extend side  $\overline{AB}$  beyond  $B$  to a point  $B'$  so that  $BB' = 3AB$ . Similarly, extend side  $\overline{BC}$  beyond  $C$  to a point  $C'$  so that  $CC' = 3BC$ , and extend side  $\overline{CA}$  beyond  $A$  to a point  $A'$  so that  $AA' = 3CA$ . The ratio of the area of  $\triangle A'B'C'$  to the area of  $\triangle ABC$  is  $Q : 1$ . What is  $Q$ ?
61. In  $\triangle ABC$ ,  $AB = 6$ ,  $BC = 7$ , and  $CA = 8$ . Point  $D$  lies on  $\overline{BC}$ , and  $\overline{AD}$  bisects  $\angle BAC$ . Point  $E$  lies on  $\overline{AC}$ , and  $\overline{BE}$  bisects  $\angle ABC$ . The bisectors intersect at  $F$ . The ratio of  $AF : FD$  is  $m : n$ . What is  $m^2 + n^2$ ?



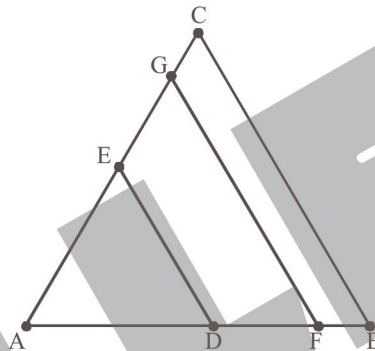
62. In  $\triangle ABC$  shown in the figure,  $AB = 7$ ,  $BC = 8$ ,  $CA = 9$ , and  $\overline{AH}$  is an altitude. Points  $D$  and  $E$  lie on sides  $\overline{AC}$  and  $\overline{AB}$ , respectively, so that  $\overline{BD}$  and  $\overline{CE}$  are angle bisectors, intersecting  $\overline{AH}$  at  $Q$  and  $P$ , respectively. If  $PQ = \frac{a}{b}\sqrt{c}$ , where  $a, b$  are co-prime, then what is the value of  $(a + b + c^2)$ ?



63. There are  $f$  noncongruent integer-sided triangles with positive area and perimeter less than 15, which are neither equilateral, isosceles, nor right triangles. What is  $f$ ?
64. In  $\triangle BAC$ ,  $\angle BAC = 40^\circ$ ,  $AB = 10$ , and  $AC = 6$ . Points  $D$  and  $E$  lie on  $\overline{AB}$  and  $\overline{AC}$  respectively. What is the minimum possible value of  $BE + DE + CD$ ?
65. In  $\triangle ABC$ ,  $AB = AC = 28$  and  $BC = 20$ . Points  $D, E$  and  $F$  are on sides  $\overline{AB}$ ,  $\overline{BC}$ , and  $\overline{AC}$ , respectively, such that  $\overline{DE}$  and  $\overline{EF}$  are parallel to  $\overline{AC}$  and  $\overline{AB}$ , respectively. What is the perimeter of parallelogram  $ADEF$ ?

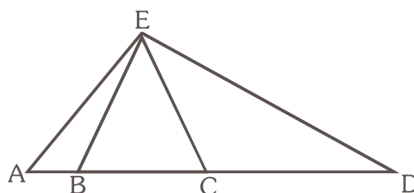


66. Triangle ABC is equilateral with  $AB = 1$ . Points E and G are on  $\overline{AC}$  and points D and F are on  $\overline{AB}$  such that both  $\overline{DE}$  and  $\overline{FG}$  are parallel to  $\overline{BC}$ . Furthermore, triangle ADE and trapezoids DFGE and FBCG all have the same perimeter.  $DE + FG$  can be written as  $\frac{r}{s}$  as  $r$  and  $s$  are co-prime. What is  $r + s$ ?

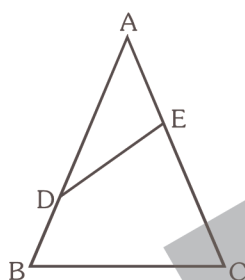


67. The angles in a particular triangle are in arithmetic progression, and the side lengths are 4, 5,  $x$ . The sum of the possible values of  $x$  equals  $a + \sqrt{b} + \sqrt{c}$  where  $a$ ,  $b$ , and  $c$  are positive integers. What is  $a + b + c$ ?
68. In triangle ABC,  $AB = 13$ ,  $BC = 14$  and  $CA = 15$ . Distinct points D, E, and F lie on segments  $\overline{BC}$ ,  $\overline{CA}$  and  $\overline{DE}$ , respectively, such that  $\overline{AD} \perp \overline{BC}$ ,  $\overline{DE} \perp \overline{AC}$ , and  $\overline{AF} \perp \overline{BF}$ . The length of segment  $\overline{DF}$  can be written as  $\frac{m}{n}$ , where  $m$  and  $n$  are relatively prime positive integers. What is  $m + n$ ?
69. Triangle ABC has  $AB = 27$ ,  $AC = 26$ , and  $BC = 25$ . Let  $I$  denote the intersection of the internal angle bisectors of  $\triangle ABC$ . What is  $BI$ ?
70. Triangle ABC has side-lengths  $AB = 12$ ,  $BC = 24$ , and  $AC = 18$ . The line through the incenter of  $\triangle ABC$  parallel to  $\overline{BC}$  intersects  $\overline{AB}$  at  $M$  and  $\overline{AC}$  at  $N$ . What is the perimeter of  $\triangle AMN$ ?
71. Points B and C lie on  $\overline{AD}$ . The length of  $\overline{AB}$  is 4 times the length of  $\overline{BD}$ , and the length of  $\overline{AC}$  is 9 times the length of  $\overline{CD}$ . The length of  $\overline{BC} \times S =$  the length of  $\overline{AD}$ . What is  $S$ ?
72. Point P is inside equilateral  $\triangle ABC$ . Points Q, R and S are the feet of the perpendiculars from P to  $\overline{AB}$ ,  $\overline{BC}$  and  $\overline{CA}$ , respectively. Given that  $PQ = 1$ ,  $PR = 2$ , and  $PS = 3$ , what is  $AB^2$ ?

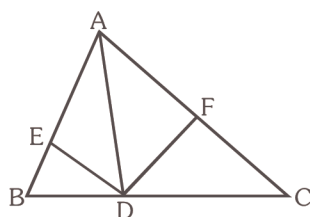
73. In triangle ABC,  $AB = (b^2 - 1)$  cm,  $BC = a^2$  cm and  $AC = 2a$  cm, where  $a$  and  $b$  are positive integers greater than 1. Find the value of  $a - b$ .
74. In the figure below, ADE is a triangle with  $\angle AED = 120^\circ$ , and B and C are points on side AD such that BCE is an equilateral triangle. If  $AB = 4$  cm,  $CD = 16$  cm and  $BC = x$  cm, find the value of  $x$ .



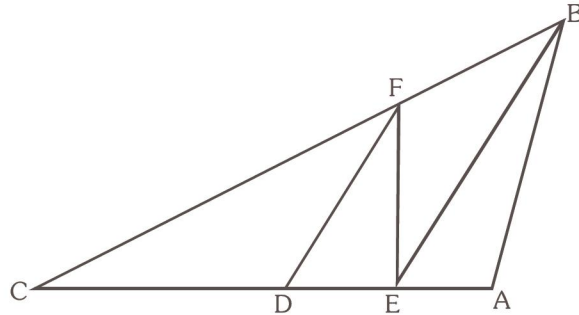
75. The figure below shows a triangle ABC where  $AB = AC$ . D and E are points on sides AB and AC, respectively, such that  $AB = 4DB$  and  $AC = 4AE$ . If the area of the quadrilateral BCED is  $52 \text{ cm}^2$  and the area of the triangle ADE is  $x \text{ cm}^2$ , find  $x$ .



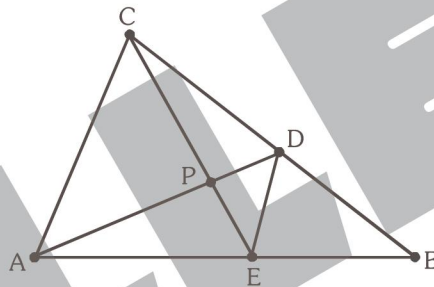
76. In the triangle ABC,  $AC = 2BC$ ,  $\angle C = 90^\circ$  and D is the foot of the altitude from C onto AB. A circle with diameter AD intersects the segment AC at E. Find  $AE : EC$ .
77. In the triangle ABC,  $AB = 8$ ,  $BC = 7$  and  $CA = 6$ . Let E be the point on BC such that  $\angle BAE = 3\angle EAC$ . Find  $\frac{4AE^2}{5}$ .
78. In the triangle ABC, the bisectors of  $\angle A$  and  $\angle B$  meet at the incentre I, the extension of AI meets the circumcircle of triangle ABC at D. Let P be the foot of the perpendicular from B onto AD, and Q a point on the extension of AD such that  $ID = DQ$ . Determine the value of  $(BQ \times IB)/(BP \times ID)$ .
79. Three sides OAB, OAC and OBC of a tetrahedron OABC are right-angled triangles, i.e.  $\angle AOB = \angle AOC = \angle BOC = 90^\circ$ . Given that  $OA = 7$ ,  $OB = 2$  and  $OC = 6$ , find the value of  $[(\text{Area of } \triangle OAB)^2 + (\text{Area of } \triangle OAC)^2 + (\text{Area of } \triangle OBC)^2 + (\text{Area of } \triangle ABC)^2 + 37]^{1/2}$ .
80. In a triangle ABC, the length of the altitudes AD and BE are 4 and 12 respectively. Find the largest possible integer value for the length of the third altitude CF.
81. In the figure below, ABC is a triangle and D is a point on side BC. Point E is on side AB such that DE is the angle bisector of  $\angle ADB$ , and point F is on side AC such that DF is the angle bisector of  $\angle ADC$ . Find the value of  $\frac{AE}{EB} \cdot \frac{BD}{DC} \cdot \frac{CF}{FA}$ .



82. In the figure below,  $ABC$  is a triangle with  $AB = 10$  cm and  $BC = 40$  cm. Points  $D$  and  $E$  lie on side  $AC$  and point  $F$  on side  $BC$  such that  $EF$  is parallel to  $AB$  and  $DF$  is parallel to  $EB$ . Given that  $BE$  is an angle bisector of  $\angle ABC$  and that  $AD = 13.5$  cm, find the length of  $CD$  in cm.

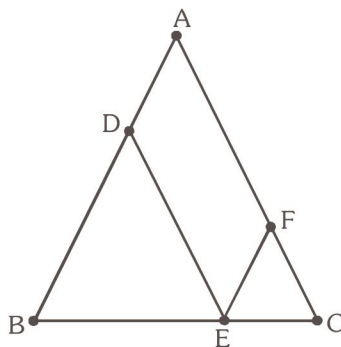


83. In triangle  $ABC$ ,  $AB = 13$ ,  $BC = 14$  and  $CA = 15$ . Distinct points  $D$ ,  $E$  and  $F$  lie on segments  $\overline{BC}$ ,  $\overline{CA}$  and  $\overline{DE}$ , respectively, such that  $\overline{AD} \perp \overline{BC}$ ,  $\overline{DE} \perp \overline{AC}$  and  $\overline{AF} \perp \overline{BF}$ . The length of segment  $\overline{DF}$  can be written as  $\frac{m}{n}$ , where  $m$  and  $n$  are relatively prime positive integers. What is  $m + n$ ?
84. In  $\triangle ABC$ , medians  $\overline{AD}$  and  $\overline{CE}$  intersect at  $P$ ,  $PE = 1.5$ ,  $PD = 2$  and  $DE = 2.5$ . What is the area of  $2[AEDC]$ ?

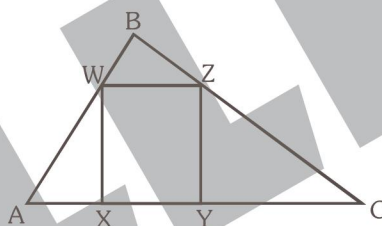


85. Triangle  $ABC$  has  $AB = 2 \cdot AC$ . Let  $D$  and  $E$  be on  $\overline{AB}$  and  $\overline{BC}$ , respectively, such that  $\angle BAE = \angle ACD$ . Let  $F$  be the intersection of segments  $AE$  and  $CD$ , and suppose that  $\triangle CFE$  is equilateral. What is  $\angle ACB$  (in degrees)?
86. A flagpole is originally 5 meters tall. A hurricane snaps the flagpole at a point  $x$  meters above the ground so that the upper part, still attached to the stump, touches the ground 1 meter away from the base. What is  $[x]$  where  $[\cdot]$  is the greatest integer function?
87. Distinct points  $A$ ,  $B$ ,  $C$ , and  $D$  lie on a line, with  $AB = BC = CD = 1$ . Points  $E$  and  $F$  lie on a second line, parallel to the first, with  $EF = 1$ . A triangle with positive area has three of the six points as its vertices. How many possible values are there for the area of the triangle?
88. An equilateral triangle has side length 6. What is the area of the region containing all points that are outside the triangle and not more than 3 units from a point of the triangle to the nearest integer? (Take  $\pi = 3.14$ )
89. Two equilateral triangles are contained in a square whose side length is  $2\sqrt{3}$ . The bases of these triangles are the opposite sides of the square and their intersection is a rhombus. If the area of the rhombus is  $8\sqrt{k} - 12$ , find  $k$ .
90. A right triangle has perimeter 32 and area 20. What is the length of its hypotenuse to the nearest integer?
91. A triangle  $\triangle ABC$  is inscribed in a circle of radius 1, with  $\angle BAC = 60^\circ$ . Altitudes  $AD$  and  $BE$  of  $\triangle ABC$  intersect at  $H$ . Find the smallest possible value of the length of the segment  $AH$ .
92. Two sides of a triangle have lengths 10 and 15. The length of the altitude to the third side is the average of the lengths of the altitudes to the two given sides. How long is the third side?

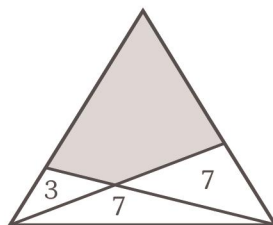
93. In  $\triangle ABC$ ,  $AB = AC = 28$  and  $BC = 20$ . Points  $D$ ,  $E$  and  $F$  are on sides  $\overline{AB}$ ,  $\overline{BC}$ , and  $\overline{AC}$  respectively, such that  $\overline{DE}$  and  $\overline{EF}$  are parallel to  $\overline{AC}$  and  $\overline{AB}$ , respectively. What is the perimeter of parallelogram  $ADEF$ ?



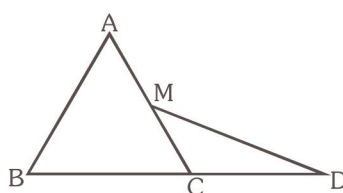
94. In  $\triangle ABC$ ,  $\angle C = 90^\circ$  and  $AB = 12$ . Squares  $ABXY$  and  $ACWZ$  are constructed outside of the triangle. The points  $X$ ,  $Y$ ,  $Z$  and  $W$  lie on a circle. The perimeter of the triangle can be written as  $a(1 + \sqrt{b})$ . Find the product of  $a$  and  $b$ .
95. Point  $P$  is inside equilateral  $\triangle ABC$ . Points  $Q$ ,  $R$  and  $S$  are the feet of the perpendiculars from  $P$  to  $\overline{AB}$ ,  $\overline{BC}$  and  $\overline{CA}$ , respectively. Given that  $PQ = 1$ ,  $PR = 2$ , and  $PS = 3$ , what is  $AB^2$ ?
96. Right  $\triangle ABC$  has  $AB = 3$ ,  $BC = 4$ , and  $AC = 5$ . Square  $XYZW$  is inscribed in  $\triangle ABC$  with  $X$  and  $Y$  on  $\overline{AC}$ ,  $W$  on  $\overline{AB}$ , and  $Z$  on  $\overline{BC}$ . What is the side length of the square to the nearest integer?



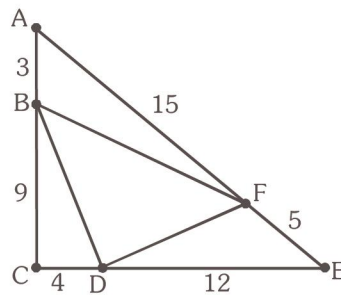
97. In a triangle with integer side lengths, one side is three times as long as a second side, and the length of the third side is 15. What is the greatest possible perimeter of the triangle?
98. A triangle is partitioned into three triangles and a quadrilateral by drawing two lines from vertices to their opposite sides. The areas of the three triangles are 3, 7, and 7, as shown. What is the area of the shaded quadrilateral?



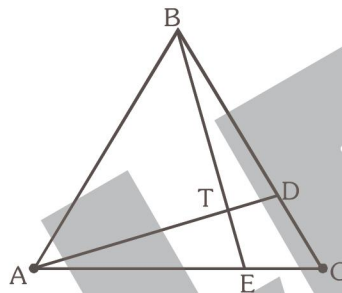
99. Equilateral  $\triangle ABC$  has side length 2,  $M$  is the midpoint of  $\overline{AC}$ , and  $C$  is the midpoint of  $\overline{BD}$ . If the area of  $\triangle CDM$  is 'x', then find the value of  $4x^2$



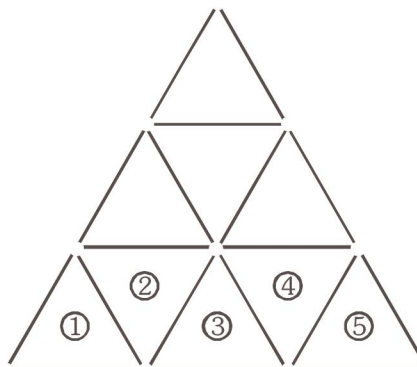
- 100.** In right triangle  $\triangle ACE$ , we have  $AC = 12$ ,  $CE = 16$ , and  $EA = 20$ . Points  $B$ ,  $D$ , and  $F$  are located on  $\overline{AC}$ ,  $\overline{CE}$ , and  $\overline{EA}$ , respectively, so that  $AB = 3$ ,  $CD = 4$ , and  $EF = 5$ . If the ratio of the area of  $\triangle ACE$  to that of  $\triangle DBF$  is  $m : n$ , find  $m + n$



- 101.** In  $\triangle ABC$  points  $D$  and  $E$  lie on  $\overline{BC}$  and  $\overline{AC}$ , respectively. If  $\overline{AD}$  and  $\overline{BE}$  intersect at  $T$  then  $AT/DT = 3$  and  $BT/ET = 4$ . What is  $\left(\frac{11CD}{BD}\right)$ ?

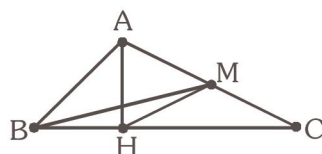


- 102.** A triangle with sides of 5, 12, and 13 has both an inscribed and a circumscribed circle. Find square of twice the distance between the centres of those circles?
- 103.** A large equilateral triangle is constructed by using toothpicks to create rows of small equilateral triangles. For example, in the figure we have 3 rows of small congruent equilateral triangles, with 5 small triangles in the base row. Total  $N$  number of toothpicks would be needed to construct a large equilateral triangle if the base row of the triangle consists of 2003 small equilateral triangles. Then find the sum of all digits of  $N$ .



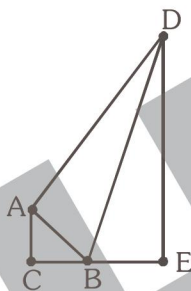
- 104.** The sides of a triangle have lengths of 15, 20, and 25. Find the length of the shortest altitude.
- 105.** Points  $A, B, C, D, E$  and  $F$  lie, in that order, on  $\overline{AF}$ , dividing it into five segments, each of length 1. Point  $G$  is not on line  $AF$ . Point  $H$  lies on  $\overline{GD}$ , and point  $J$  lies on  $\overline{GF}$ . The line segments  $\overline{HC}$ ,  $\overline{JE}$ , and  $\overline{AG}$  are parallel. Find  $\left(\frac{3HC}{JE}\right)$ .

- 106.** Points A, B, C and D lie on a line, in that order, with  $AB = CD$  and  $BC = 12$ . Point E is not on the line, and  $BE = CE = 10$ . The perimeter of  $\triangle AED$  is twice the perimeter of  $\triangle BEC$ . Find AB.
- 107.** Let  $\triangle XOY$  be a right-angled triangle with  $m\angle XOY = 90^\circ$ . Let M and N be the midpoints of legs OX and OY, respectively. Given that  $XN = 19$  and  $YM = 22$ , find XY.
- 108.** In  $\triangle ABC$ ,  $\angle A = 100^\circ$ ,  $\angle B = 50^\circ$ ,  $\angle C = 30^\circ$ ,  $\overline{AH}$  is an altitude and  $\overline{BM}$  is a median. Then  $\angle MHC$  (in degrees) is

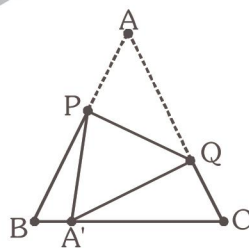


- 109.** Triangle ABC has a right angle at C,  $AC = 3$  and  $BC = 4$ . Triangle ABD has a right angle at A and  $AD = 12$ . Points C and D are on opposite sides of  $\overline{AB}$ . The line through D parallel to  $\overline{AC}$  meets  $\overline{CB}$  extended at E. If

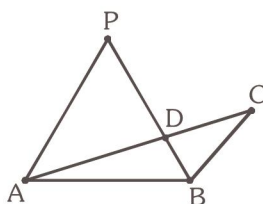
$$\frac{DE}{DB} = \frac{m}{n} \text{ where } m \text{ and } n \text{ are relatively prime positive integers, then } \frac{m+n}{2} \text{ is}$$



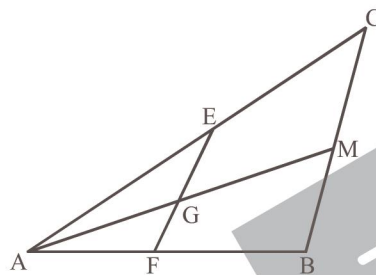
- 110.** Equilateral triangle ABC has been creased and folded so that vertex A now rests at  $A'$  on BC as shown. If  $BA' = 1$  and  $A'C = 2$ . If the length of crease PQ is  $\frac{k\sqrt{21}}{20}$ , find k.



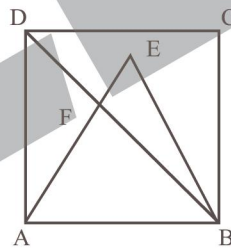
- 111.** In triangle ABC,  $\angle ABC = 120^\circ$ ,  $AB = 3$  and  $BC = 4$ . If perpendiculars constructed to  $\overline{AB}$  at A and to  $\overline{BC}$  at C meet at D, then  $\sqrt{3}CD$  equals?
- 112.** Triangle ABC and point P in the same plane are given. Point P is equidistant from A and B, angle APB is twice angle ACB, and  $\overline{AC}$  intersects  $\overline{BP}$  at point D. If  $PB = 3$  and  $PD = 2$ , then  $AD \cdot CD =$



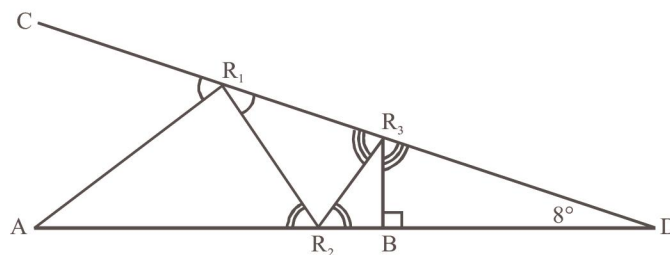
- 113.** In triangle ABC,  $AB = 13$ ,  $BC = 14$ , and  $AC = 15$ . Let D denote the midpoint of  $\overline{BC}$  and let E denote the intersection of  $\overline{BC}$  with the bisector of angle BAC. Which of the following is closest to the area of the triangle ADE?
- 114.** Two congruent  $30^\circ - 60^\circ - 90^\circ$  triangles are placed so that they overlap partly and their hypotenuses coincide. If the hypotenuse of each triangle is 12, and the area common to both triangles is given by A. Find the value of  $\sqrt{3}A$ .
- 115.** In  $\triangle ABC$  with right angle at C, altitude CH and median CM trisect the right angle. If the area of  $\triangle CHM$  is 10, then the area of  $\triangle ABC$  is
- 116.** In triangle ABC shown in the adjoining figure, M is the mid-point of side BC,  $AB = 12$  and  $AC = 16$ . Points E and F are taken on AC and AB, respectively, and lines EF and AM intersect at G. If  $AE = 2AF$  and  $EG/GF$  is X. Find the value of  $10X$ .



- 117.** Vertex E of equilateral triangle ABE is in the interior of square ABCD, and F is the point of intersection of diagonal BD and line segment AE. If length AB is  $\sqrt{1+\sqrt{3}}$  and the area of  $\triangle ABF$  is given by A. Find  $(4A)^2$ .

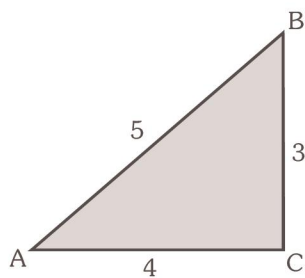


- 118.** A ray of light originates from point A and travels in a plane, being reflected n times between lines AD and CD, before striking a point B (which may be on AD or CD) perpendicularly and retracing its path to A. (At each point of reflection the light makes two equal angles as indicated in the adjoining figure. The figure shows the light path for  $n = 3$ .) If  $\angle CDA = 8^\circ$ , what is the largest value n can have?

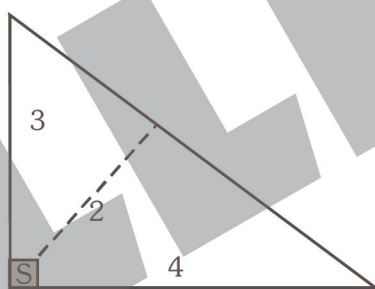


- 119.** In a triangle with side of lengths a, b and c,  $(a + b + c)(a + b - c) = 3ab$ . The measure of the angle opposite the side of length c is

120. A paper triangle with sides of lengths 3, 4, and 5 inches, as shown, is folded so that point A falls on point B. Then the length in inches of the crease is L. Find the value of  $8L$ .



121. Right triangle ABC has leg lengths  $AB = 20$  and  $BC = 21$ . Including  $\overline{AB}$  and  $\overline{BC}$ , how many line segments with integer length can be drawn from vertex B to a point on hypotenuse  $\overline{AC}$ ?
122. Farmer Pythagoras has a field in the shape of a right triangle. The right triangle's legs have lengths 3 and 4 units. In the corner where those sides meet at a right angle, he leaves a small unplanted square S so that from the air it looks like the right angle symbol. The rest of the field is planted. The shortest distance from S to the hypotenuse is 2 units. Let the fraction of the field planted can be written as  $\frac{x}{y}$ . Find the value of  $10(y - x)$ . Here x, y are co-prime numbers.



123. Triangle ABC with  $AB = 50$  and  $AC = 10$  has area 120. Let D be the midpoint of  $\overline{AB}$ , and let E be the midpoint of  $\overline{AC}$ . The angle bisector of  $\angle BAC$  intersects  $\overline{DE}$  and  $\overline{BC}$  at F and G, respectively. What is the area of quadrilateral FDBG?
124. Line segment  $\overline{AB}$  is a diameter of a circle with  $AB = 24$ . Point C, not equal to A or B, lies on the circle. As point C moves around the circle, the centroid (center of mass) of  $\triangle ABC$  traces out a closed curve missing two points. To the nearest positive integer, what is the area of the region bounded by this curve?
125. A square with side length x is inscribed in a right triangle with sides of length 3, 4, and 5 so that one vertex of the square coincides with the right-angle vertex of the triangle. A square with side length y is inscribed in another right angle triangle with sides of length 3, 4 and 5 so that one side of the square lies on the hypotenuse of the triangle. Find the value of  $(7x + 37y)$ .
126. Rectangle ABCD has  $AB = 3$  and  $BC = 4$ . Point E is the foot of the perpendicular from B to diagonal  $\overline{AC}$ . If the area of  $\triangle ADE$  is given by P. Find  $25P$ .

- 127.** Let  $ABC$  be an equilateral triangle. Extend side  $\overline{AB}$  beyond  $B$  to a point  $B'$  so that  $BB' = 3AB$ . Similarly, extend side  $\overline{BC}$  beyond  $C$  to a point  $C'$  so that  $CC' = 3BC$ , and extend side  $\overline{CA}$  beyond  $A$  to a point  $A'$  so that  $AA' = 3CA$ . If the ratio of the area of  $\triangle A'B'C'$  to the area of  $\triangle ABC$  is  $x : 1$ . Find the value of  $x$ .
- 128.** A thin piece of wood of uniform density in the shape of an equilateral triangle with side length 3 inches weighs 12 gm. A second piece of the same type of wood, with the same thickness, also in the shape of an equilateral triangle, has side length of 5 inches. If  $W$  is the weight, in gm, of the second piece. Then find  $[W]$ , where  $[x]$  denotes the greatest integer less than or equal to  $x$ .
- 129.** The isosceles right triangle  $ABC$  has right angle at  $C$  and area 12.5. The rays trisecting  $\angle ACB$  intersect  $AB$  at  $D$  and  $E$ . If the area of  $\triangle CDE$  can be written as  $\frac{a(2-\sqrt{b})}{2}$ . Find the product of  $a$  and  $b$ .

ALLEN

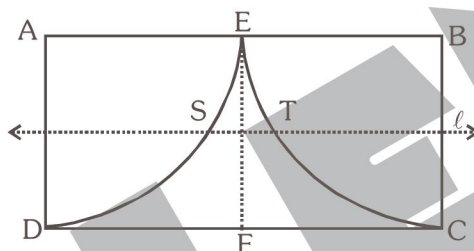
## Important Notes

ALLEN

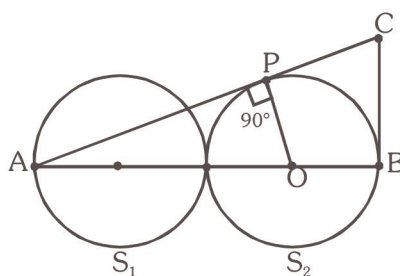
## 4.2 Circle geometry

### EXERCISE - 4.2

- A circle is inscribed in a triangle ABC. It touches AB at D.  $AD = 5$  cm,  $DB = 3$  cm and  $\angle A = 60^\circ$ . Then BC is  
 (1) 8 cm (2) 13 cm (3) 20 cm (4) 31 cm
- ABC is a triangle  $AB = 6$  cm,  $BC = 9$  cm,  $CA = 7$  cm. Circles are drawn with centres at A, B, C. The circles with centres A and B touch externally. The circle with centre C touches these two circles internally. The sum of the radii of these circles (in cm) is  
 (1) 22 (2) 17 (3) 44 (4) 111
- In rectangle ABCD,  $AB = 2BC = 4$  cm E and F are midpoints of AB and CD respectively. ESD and ETC are arcs of circles centred at A and B respectively. If the perpendicular bisector line  $\ell$  of EF cuts the arcs at S and T as in the diagram, then ST is equal to (in cm)

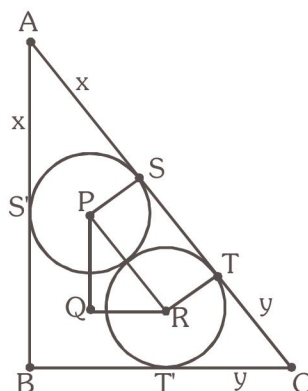


- (1)  $(4 - 2\sqrt{3})$  (2)  $(3 + \sqrt{3})$  (3)  $(2 + 2\sqrt{3})$  (4)  $(4\sqrt{3} - 2)$
- An isosceles trapezoid is circumscribed about a circle of radius 2 cm and the area of the trapezoid is  $20\text{ cm}^2$ . The equal sides of the trapezoid have length equal to?  
 (1) 1 (2) 5 (3) 2 (4) 3
- ABC and ADE are two secants of a circle of radius 3 cm. A is at a distance of 5 cm from the centre of the circle. The secants include an angle of  $30^\circ$ . The area of the  $\triangle ACE$  is  $10\text{ cm}^2$ . Then the area of the  $\triangle ADB$  (in  $\text{cm}^2$ ) is \_\_\_\_.  
 (1)  $\frac{8}{5}$  (2)  $\frac{7}{5}$  (3)  $\frac{6}{5}$  (4)  $\frac{9}{5}$
- In the figure below two equal circles  $S_1, S_2$  of radii 2 units each touch each other. AB is the common diameter. The tangent at B meets the tangent from A to the circle  $S_2$  at C as shown. If  $BC = k\sqrt{2}$  then the value of K is.

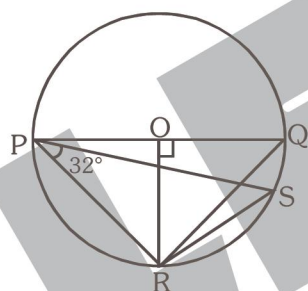


- (1) 2 (2) 2.2 (3) 3 (4) 4

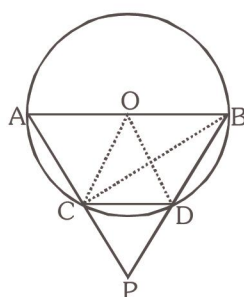
7. In the figure below, the triangle is a 3 – 4 – 5 sides triangle. Two equal circles are placed as in the figure. The radius of each circle is



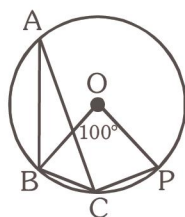
- (1)  $\frac{4}{5}$  (2)  $\frac{5}{8}$  (3)  $\frac{6}{11}$  (4)  $\frac{5}{7}$
8. In figure, PQ is a diameter of a circle with centre at O and  $OR \perp PQ$ , where R is a point on the circle. If S is another point on the circle such that  $\angle RPS = 32^\circ$ , then  $\angle QRS$  is



- (1)  $13^\circ$  (2)  $26^\circ$  (3)  $45^\circ$  (4) None of these
9. In the given figure, AB is the diameter of a circle C (O, r). Chord CD is equal to radius OC. If AC and BD intersect at P, find  $\angle APB$ .

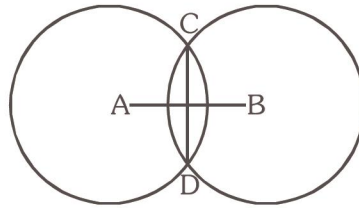


- (1)  $30^\circ$  (2)  $45^\circ$  (3)  $60^\circ$  (4)  $90^\circ$
10. In the given figure,  $\angle BCP$  is equal to

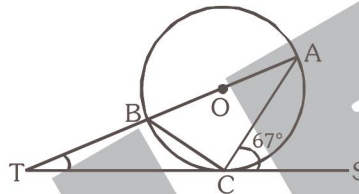


- (1)  $60^\circ$  (2)  $100^\circ$  (3)  $50^\circ$  (4)  $130^\circ$

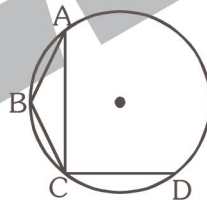
11. In the figure given below, A and B are the centers of the two congruent circles with radius 17 units. If  $AB = 30$  units, the length of the common chord DC is :



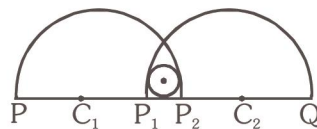
- (1) 25 units                      (2) 10 units                      (3) 10 units                      (4) 16 units
12. Two circles with centres A and B have radii 8 cm and 1 cm respectively. The distance AB is 13 cm. A third circle with centre C and radius  $r$  cm touches both circles externally. If angle ACB is  $90^\circ$ , then value of  $r$  is:
- (1) 11                      (2) 4                      (3) 5                      (4) 2
13. Two circles of radii 2 and 3 cm touch each other externally. The length of direct common tangent to the two circles will be :
- (1)  $2\sqrt{6}$  cm                      (2)  $\sqrt{26}$  cm                      (3) 5 cm                      (4) 2.4 cm
14. As shown in the figure, if O is the centre of the circle, then find the measure of the angle  $\angle BTC$ .



- (1)  $23^\circ$                       (2)  $67^\circ$                       (3)  $51^\circ$                       (4)  $44^\circ$
15. In the following figure,  $AB = BC$ ,  $AC = CD$  and  $\angle ACD = 90^\circ$ . If the radius of the circle is " $r$ " units then find the length of the chord BC.



- (1)  $r\sqrt{\sqrt{2}-1}$                       (2)  $r\sqrt{\sqrt{3}-1}$                       (3)  $r\sqrt{2-\sqrt{2}}$                       (4)  $r\sqrt{2-\sqrt{3}}$
16.  $C_1$  and  $C_3$  are centres of two identical semi circles with radius ' $r$ ' units. The ratio of  $C_1P_1$  to PQ is 1 : 7, as shown in the figure. Find the radius of the circle inscribed in the region common to the identical semicircles, given that the circle touches the two semicircles and the line segment  $P_1P_2$ .

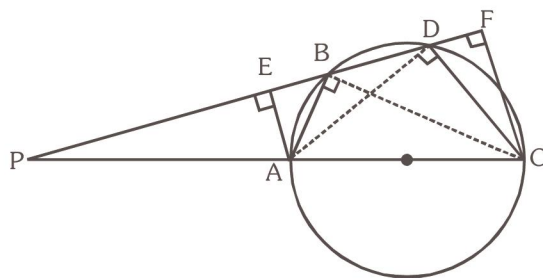


- (1)  $\frac{7r}{32}$                       (2)  $\frac{3r}{16}$                       (3)  $\frac{r}{4}$                       (4)  $\frac{5r}{16}$
17. ABC is triangle with side lengths 13, 14 and 15 units. If I is the incentre and R is its circum radius, then the value of  $\frac{AI \times BI \times CI}{R}$  is equal to (in Sq. units)
- (1) 130                      (2) 64                      (3) 126                      (4) 100

18. Two adjacent vertices of a square are on a circle of radius  $R$  and the other two vertices lie on a tangent to the circle. The length of the side of the square is

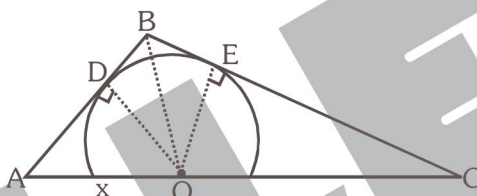
(1)  $\frac{4R}{3}$  (2)  $\frac{6R}{5}$  (3)  $\frac{8R}{5}$  (4)  $\frac{3R}{2}$

19. A circle with diameter  $\overline{AC}$  is intersected by a secant at point  $B$  and  $D$ . The secant and the diameter intersect at point  $P$  outside the circle. Perpendiculars  $\overline{AE}$  and  $\overline{CF}$  are drawn from the extremities of the diameter to the secant. If  $EB = 2$ , and  $BD = 6$ , find  $DF$ .



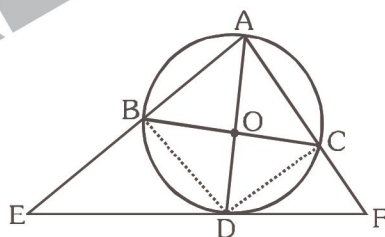
(1) 2 (2) 3 (3) 4 (4) 5

20. In  $\triangle ABC$ , in which  $AB = 12$ ,  $BC = 18$ , and  $AC = 25$ , a semicircle is drawn so that diameter lies on  $\overline{AC}$ , and so that it is tangent to  $\overline{AB}$  and  $\overline{BC}$ . If  $O$  is the center of the circle, find the measure of  $\overline{AO}$ .



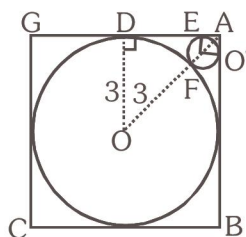
(1) 10 (2) 20 (3) 30 (4) 40

21. Triangle  $ABC$  is inscribed in a circle with diameter  $\overline{AD}$ . A tangent to the circle at  $D$  cuts  $\overline{AB}$  extended at  $E$  and  $\overline{AC}$  extended at  $F$ . If  $AB = 4$ ,  $AC = 6$ , and  $BE = 8$ , find  $CF$ .



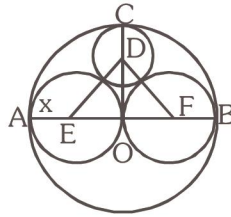
(1) 1 (2) 2 (3) 4 (4) 5

22. A circle with radius 3 is inscribed in a square. Find the radius of the circle that is inscribed between two sides of the square and the original circle.



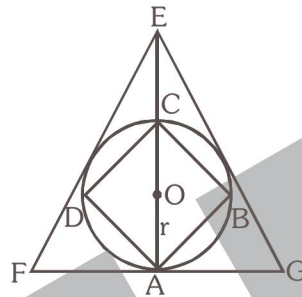
(1)  $3(3 + 2\sqrt{2})$  (2)  $3(3 - 2\sqrt{2})$  (3)  $3(2 + 2\sqrt{2})$  (4)  $3(2 - 2\sqrt{2})$

23.  $\overline{AB}$  is diameter of circle O. Two circles are drawn with  $\overline{AO}$  and  $\overline{OB}$  as diameters. In the region between the circumferences, a circle D is inscribed tangent to the three previous circles. If the measure of the radius of circle D is 8, find AB.



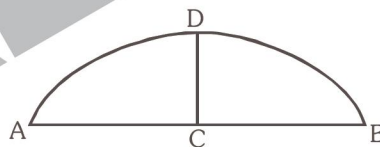
- (1) 48                      (2) 50                      (3) 52                      (4) 54

24. Find the ratio between the area of a square inscribed in a circle and an equilateral triangle circumscribed about the same circle.

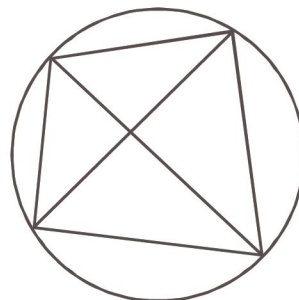


- (1)  $\frac{2\sqrt{9}}{9}$                       (2)  $\frac{2\sqrt{3}}{9}$                       (3)  $\frac{2\sqrt{5}}{9}$                       (4)  $\frac{2\sqrt{6}}{9}$

25. The diagram shows a segment of a circle such that CD is perpendicular bisector of the chord AB. Given that  $AB = 16$  and  $CD = 4$ , find the diameter of the circle.

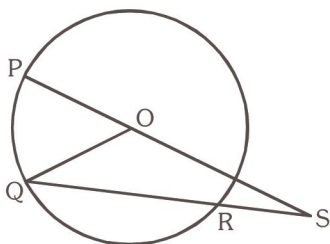


26. n dots are drawn on the circumference of a circle. By joining all the dots to one another by straight lines, the maximum number of regions that can be formed in the circle is counted. For example, when  $n = 4$  the maximum number of regions is 8.

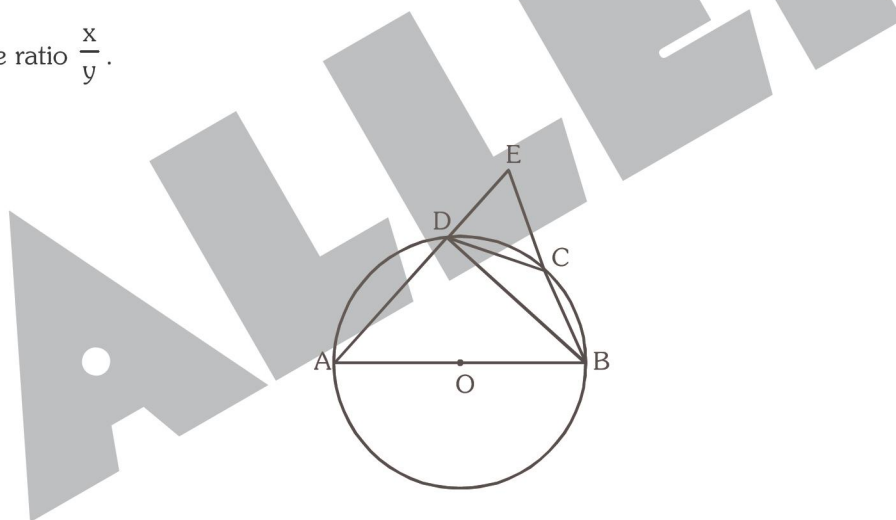


What is the maximum number of regions that can be formed when  $n = 7$ ?

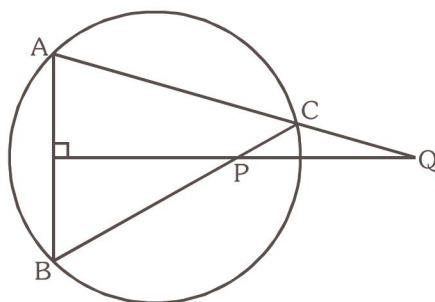
- 27.** In the diagram, P, Q and R are three points on the circle whose centre is O. The lines PO and QR are produced to meet at S. Suppose that  $RS = OP$ , and  $\angle PSQ = 12^\circ$  and  $\angle POQ = x^\circ$ . Find the value of  $x$ .



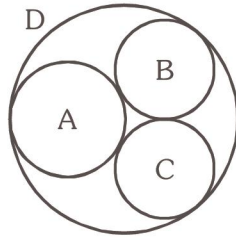
- 28.**  $\triangle ABC$  is a triangle such that  $\angle C = 90^\circ$ . Suppose  $AC = 156$  cm,  $AB = 169$  cm and the perpendicular distance from  $C$  to  $AB$  is  $x$  cm. Find the value of  $x$ .
- 29.** Circles  $C_1$  and  $C_2$  have radii 3 and 7 respectively. The circles intersect at distinct points  $A$  and  $B$ . A point  $P$  outside  $C_2$  lies on the line determined by  $A$  and  $B$  at a distance of 5 from the center of  $C_1$ . Point  $Q$  is chosen on  $C_2$  so that  $PQ$  is tangent to  $C_2$  at  $Q$ . Find the length of the segment  $PQ$ .
- 30.** Let  $C$  be a circle with radius 2006. Suppose  $n$  points are placed inside the circle and the distance between any two points exceed 2006. What is the largest possible  $n$ ?
- 31.** In the following figure,  $AB$  is the diameter of a circle with centre at  $O$ . It is given that  $AB = 4$  cm,  $BC = 3$  cm,  $\angle ABD = \angle DBE$ . Suppose the area of the quadrilateral  $ABCD$  is  $x$  cm<sup>2</sup> and the area of  $\triangle DCE$  is  $y$  cm<sup>2</sup>. Find the value of the ratio  $\frac{x}{y}$ .



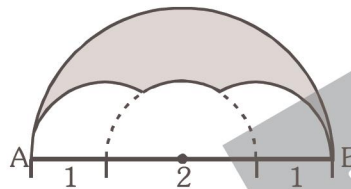
- 32.** Points A, B, C lie on a circle centered at O with radius 7. The perpendicular bisector of AB meets the segment BC at P and the extension of AC at Q. Determine the value of  $OP \cdot OQ$ .



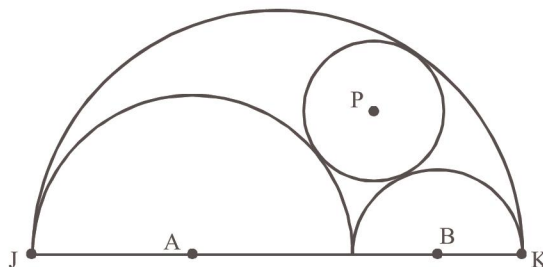
33. Circles A, B and C are externally tangent to each other and internally tangent to circle D. Circles B and C are congruent. Circle A has radius 1 and passes through the center of D. If the radius of circle B is  $\frac{a}{b}$  then the sum of a and b is?



34. Three semicircles of radius 1 are constructed on diameter AB of a semicircle of radius 2. The centers of the small semicircles divide  $\overline{AB}$  into four line segments of equal length, as shown. If the area of the shaded region that lies within the large semicircle but outside the smaller semicircles is  $\frac{7\pi - a\sqrt{3}}{6}$ , then value of a = ?

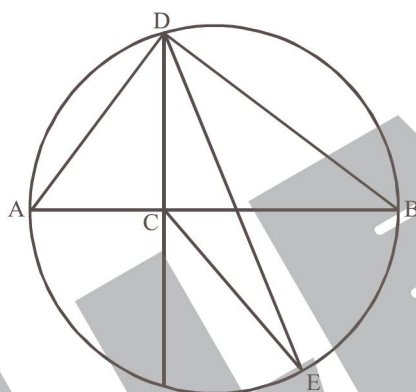


35. In  $\triangle ABC$  with side lengths  $AB = 13$ ,  $AC = 12$ , and  $BC = 5$ , let O and I denote the circumcenter and incenter, respectively. A circle with center M is tangent to the legs AC and BC and to the circumcircle of  $\triangle ABC$ . The area of  $\triangle MOI$  can be written as  $\frac{p}{q}$  in simplest form. What is  $p^2 + q^2$ ?
36. In the figure below, semicircles with centers at A and B and with radii 2 and 1, respectively, are drawn in the interior of, and sharing bases with, a semicircle with diameter  $\overline{JK}$ . The two smaller semicircles are externally tangent to each other and internally tangent to the largest semicircle. A circle centered at P is drawn externally tangent to the two smaller semicircles and internally tangent to the largest semicircle. Radius of the circle centered at P is  $\frac{a}{b}$  where a and b are coprime. What is  $5a + 6b$ ?

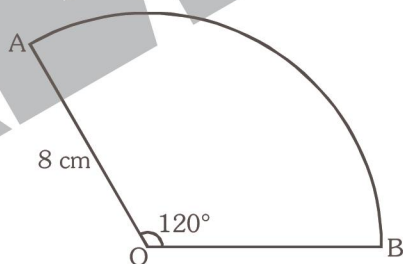


37. The diameter  $\overline{AB}$  of a circle of radius 2 is extended to a point D outside the circle so that  $BD = 3$ . Point E is chosen so that  $ED = 5$  and the line ED is perpendicular to the line AD. Segment  $\overline{AE}$  intersects the circle at point C between A and E. The area of  $\triangle ABC$  is  $\frac{P}{S}$  where P and S are co-prime. What is  $P - 2S$ ?

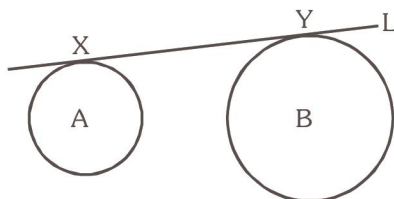
38. Circles with centers P, Q and R, having radii 1, 2 and 3, respectively, lie on the same side of line  $\ell$  and are tangent to  $\ell$  at P', Q' and R', respectively, with Q' between P' and R'. The circle with centre Q is externally tangent to each of the other two circles. The area of triangle PQR is  $\sqrt{a} - \sqrt{b}$ . What is  $a^2 + b^2$ ?
39. Quadrilateral ABCD is inscribed inside a circle with  $\angle BAC = 70^\circ$ ,  $\angle ADB = 40^\circ$ , AD = 4, and BC = 6. What is (AC)<sup>2</sup>?
40. In  $\triangle ABC$ , AB = 86, and AC = 97. A circle with center A and radius AB intersects  $\overline{BC}$  at points B and X. Moreover  $\overline{BX}$  and  $\overline{CX}$  have integer lengths. What is BC?
41. Circle  $C_1$  has its center O lying on circle  $C_2$ . The two circles meet at X and Y. Point Z in the exterior of  $C_1$  lies on circle  $C_2$  and XZ = 13, OZ = 11, and YZ = 7. The radius of circle  $C_1$  is  $\sqrt{R}$ . What is R?
42. Let  $\overline{AB}$  be a diameter of a circle and C be a point on  $\overline{AB}$  with  $2 \cdot AC = BC$ . Let D and E be points on the circle such that  $\overline{DC} \perp \overline{AB}$  and  $\overline{DE}$  is a second diameter. The ratio of the area of  $\triangle DCE$  to the area of  $\triangle ABD$  is  $\frac{21}{y}$ . What is y?



43. The diagram shows a sector OAB of a circle, centre O and radius 8 cm, in which  $\angle AOB = 120^\circ$ . Another circle of radius r cm is to be drawn through the points O, A and B. Find the value of r.

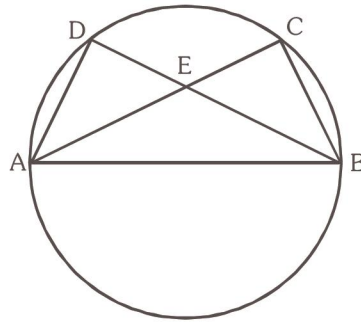


44. The figure below shows two circles with centres A and B, and a line L which is a tangent to the circles at X and Y. Suppose that XY = 40 cm, AB = 41 cm and the area of the quadrilateral ABYX is 300 cm<sup>2</sup>. If a and b denote the areas of the circles with centre A and centre B respectively, find the value of  $\frac{b}{a}$ .

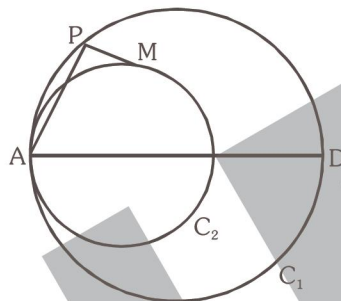


45. Let  $A_1, A_2, A_3, A_4, A_5$  and  $A_6$  be six points on a circle in this order such that  $\widehat{A_1A_2} = \widehat{A_2A_3}$ ,  $\widehat{A_3A_4} = \widehat{A_4A_5}$  and  $\widehat{A_5A_6} = \widehat{A_6A_1}$ , where  $\widehat{A_1A_2}$  denotes the arc length of the arc  $A_1A_2$  etc. It is also known that  $\angle A_1A_3A_5 = 72^\circ$ . Find the size of  $\angle A_4A_6A_2$  in degrees.

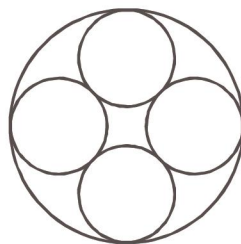
46. The figure below shows a circle with diameter  $AB$ .  $C$  and  $D$  are points on the circle on the same side of  $AB$  such that  $BD$  bisects  $\angle CBA$ . The chords  $AC$  and  $BD$  intersect at  $E$ . It is given that  $AE = 169$  cm and  $EC = 119$  cm. If  $ED = x$  cm, find the value of  $x$ .



47. Two circles  $C_1$  and  $C_2$  of radii 10 cm and 8 cm respectively are tangent to each other internally at a point  $A$ .  $AD$  is the diameter of  $C_1$  and  $P$  and  $M$  are points on  $C_1$  and  $C_2$  respectively such that  $PM$  is tangent to  $C_2$ , as shown in the figure below. If  $PM = \sqrt{20}$  cm and  $\angle PAD = x^\circ$ , find the value of  $x$ .

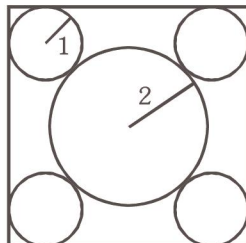


48. Let  $CD$  be a chord of a circle  $\Gamma_1$  and  $AB$  a diameter of  $\Gamma_1$  perpendicular to  $CD$  at  $N$  with  $AN > NB$ . A circle  $\Gamma_2$  centred at  $C$  with radius  $CN$  intersects  $\Gamma_1$  at points  $P$  and  $Q$ , and the segments  $PQ$  and  $CD$  intersect at  $M$ . Given that the radii of  $\Gamma_1$  and  $\Gamma_2$  are 61 and 60 respectively, find the length of  $AM$ .
49. Let  $B$  be a point on the circle centred at  $O$  with diameter  $AC$  and let  $D$  and  $E$  be the circumcentres of the triangles  $OAB$  and  $OBC$  respectively. Given that  $\sin \angle BOC = \frac{4}{5}$  and  $AC = 24$ , find the area of the triangle  $BDE$ .
50. Circle  $A$  has radius 100. Circle  $B$  has an integer radius  $r < 100$  and remains internally tangent to circle  $A$  as it rolls once around the circumference of circle  $A$ . The two circles have the same points of tangency at the beginning and end of circle  $B$ 's trip. How many possible values can  $r$  have?
51. Many cathedrals have windows with portions containing a ring of congruent circles that are circumscribed by a larger circle. In the figure shown, the number of smaller circles is four. If the ratio of the sum of the areas of the four smaller circles to the area of the larger circle is  $2[x + 4y\sqrt{2}]$  then find the value of  $x^2 + y^3$

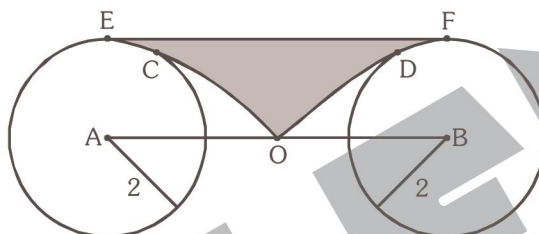


52. Points  $A$  and  $B$  lie on a circle centered at  $O$ , and  $\angle AOB = 60^\circ$ . A second circle is internally tangent to the first and tangent to both  $\overline{OA}$  and  $\overline{OB}$ . If the ratio of the area of the smaller circle to that of the larger circle is  $\frac{x}{y}$  then find the value of  $x^2 + y^2$ .

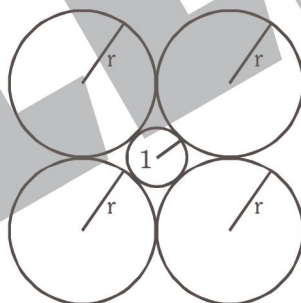
53. Points A and B are on a circle of radius 5 and  $AB = 6$ . Point C is the midpoint of the minor arc AB. What is the length of the line segment AC?
54. Four circles of radius 1 are each tangent to two sides of a square and externally tangent to a circle of radius 2, as shown. If the area of the square is  $x + y\sqrt{2}$  then find the value of  $\frac{5x}{x-y}$ ?



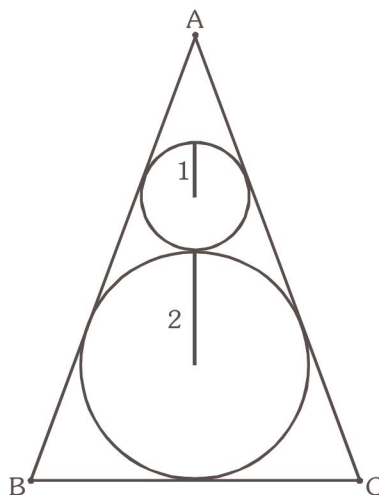
55. Circles centered at A and B each have radius 2, as shown. Point O is the midpoint of  $\overline{AB}$ , and  $OA = 2\sqrt{2}$ . Segments OC and OD are tangent to the circles centered at A and B, respectively, and EF is a common tangent. What is the area of the shaded region ECODF to the nearest integer? (Take  $\pi = 3.14$ )



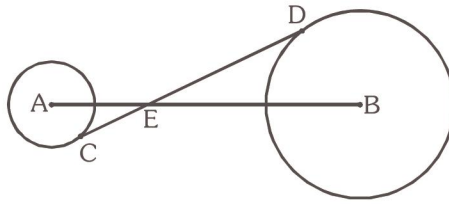
56. A circle of radius 1 is surrounded by 4 circles of radius r as shown. Find the value of r to the nearest whole number.



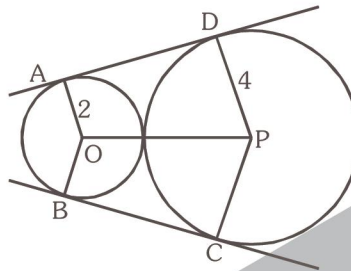
57. A circle of radius 1 is tangent to a circle of radius 2. The sides of  $\triangle ABC$  are tangent to the circles as shown, and the sides  $\overline{AB}$  and  $\overline{AC}$  are congruent. What is the area of  $\triangle ABC$  to the nearest integer?



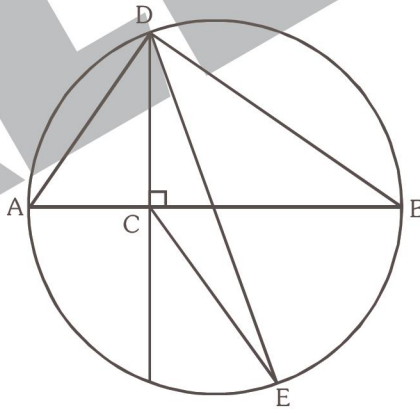
58. Circles with centers A and B have radii 3 and 8, respectively. A common internal tangent intersects the circles at C and D, respectively. Lines AB and CD intersect at E, and  $AE = 5$ . What is  $3CD$ ?



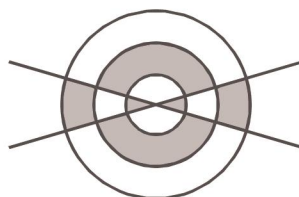
59. Circles with centers O and P have radii 2 and 4, respectively, and are externally tangent. Points A and B are on the circle centered at O, and points C and D are on the circle centered at P, such that  $\overline{AD}$  and  $\overline{BC}$  are common external tangents to the circles. If the area of hexagon AOBCPD is  $x\sqrt{3}$  then find the value of x?



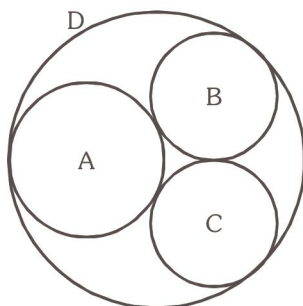
60. Let  $\overline{AB}$  be a diameter of a circle and C be a point on  $\overline{AB}$  with  $2 \cdot AC = BC$ . Let D and E be points on the circle such that  $\overline{DC} \perp \overline{AB}$  and  $\overline{DE}$  is a second diameter. If the ratio of the area of  $\triangle DCE$  to the area of  $\triangle ABD$  is  $\frac{x}{y}$  then find  $\sqrt{\frac{1}{x} + y}$



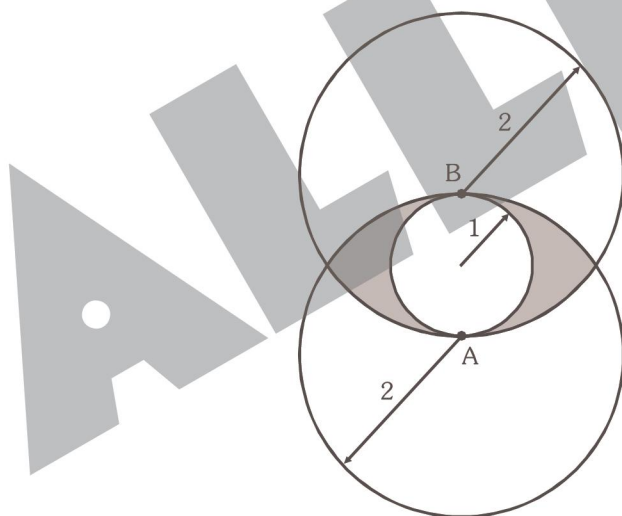
61. Two distinct lines pass through the centre of three concentric circles of radii 3, 2, and 1. The area of the shaded region in the diagram is  $\frac{8}{13}$  of the area of the unshaded region. Find  $\frac{7}{3}$  times the measure of the acute angle formed by the two lines? (Note:  $\pi$  radians is 180 degrees.)



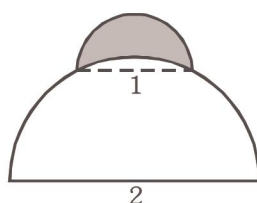
62. Circles A, B, and C are externally tangent to each other and internally tangent to circle D. Circles B and C are congruent. Circle A has radius 1 and passes through the centre of D. What is three times the square root of diameter of circle B?



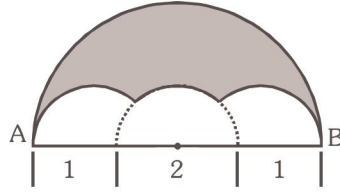
63. In  $\triangle ABC$  we have,  $AB = 7$ ,  $AC = 8$ , and  $BC = 9$ . Point D is on the circumscribed circle of the triangle so that  $\overline{AD}$  bisects  $\angle BAC$ . What is the value of  $\left(\frac{3AD}{CD}\right)$ ?
64. A circle of radius 1 is internally tangent to two circles of radius 2 at points A and B, where AB is a diameter of the smaller circle. If the area of the region, shaded in the figure, that is outside the smaller circle and inside each of the two larger circles is represented by  $\left(\frac{a\pi}{3} + b\sqrt{3}\right)$ , then find value of  $(a + b)$ .



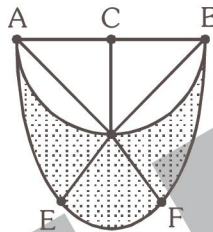
65. A semicircle of diameter 1 sits at the top of a semicircle of diameter 2, as shown. The shaded area inside the smaller semicircle and outside the larger semicircle is called a lune. If the area of this lune is in the form of  $\left(\frac{\sqrt{3}}{x} - \frac{\pi}{y}\right)$  then find  $\left(\frac{y}{x}\right)$ .



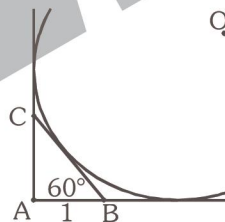
66. Three semicircles of radius 1 are constructed on diameter  $\overline{AB}$  of a semicircle of radius 2. The centers of the small semicircles divide  $\overline{AB}$  into four line segments of equal length, as shown. Then the area of the shaded region that lies within the large semicircle but outside the smaller semicircles is given by  $\left(\frac{a}{b}\pi - \frac{c}{d}\right)$ . Find the value of  $(a^2 + b^2 + c^2 + d^2)$ .



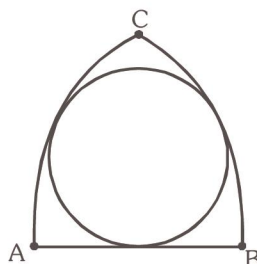
67. Semicircle  $\widehat{AB}$  has center C and radius 1. Point D is on  $\widehat{AB}$  and  $\overline{CD} \perp \overline{AB}$ . Extend  $\widehat{BD}$  and  $\widehat{AD}$  to E and F respectively, so that circular arcs  $\widehat{AE}$  and  $\widehat{BF}$  have B and A as their respective centers. circular arc EF has center D. The area of the shaded “smile”, AEFBDA is  $(2\pi - \pi\sqrt{2} - k)$ , find k.



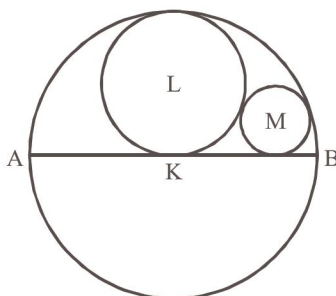
68. A circle of radius r has chord  $\overline{AB}$  length 10 and  $\overline{CD}$  of length 7. When  $\overline{AB}$  and  $\overline{CD}$  are extended through B and C, respectively, they intersect at P, which is outside the circle. If  $\angle APD = 60^\circ$  and  $BP = 8$ , then  $r^2 =$
69. A circle with centre O is tangent to the coordinate axes and to the hypotenuse of the  $30^\circ - 60^\circ - 90^\circ$  triangle ABC as shown, where  $AB = 1$ . What is the radius of the circle (round off to nearest natural number)?



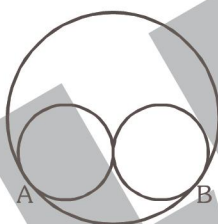
70. A circle is circumscribed about a triangle with sides 3, 4, and 5, thus dividing the interior of the circle into four regions. Let A, B, and C be the areas of the non-triangular regions, with C being the largest. Then  $-(A + B - C) = ?$
71. If circular arcs AC and BC have centers at B and A, respectively, then there exists a circle tangent to both  $\widehat{AC}$  and  $\widehat{BC}$ , and to  $\overline{AB}$ . If the length of  $\widehat{BC}$  is 12, then the circumference of the circle is



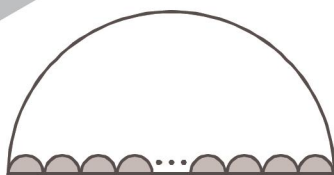
72. In the adjoining figure, circle K has diameter AB; circle L is tangent to circle K and to AB at the center of circle K; and circle M is tangent to circle K, to circle L and to AB. The ratio of the area of circle K to the area of circle M is



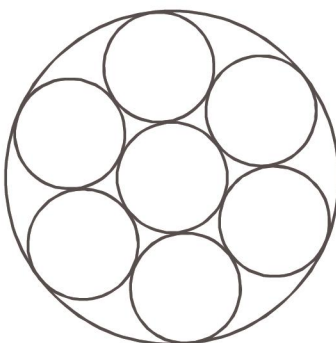
73. A circle with area  $A_1$  is contained in the interior of a larger circle with area  $A_1 + A_2$ . If the radius of the larger circle is 3, and if  $A_1, A_2, A_1 + A_2$  is an arithmetic progression, and the radius of the smaller circle is given by  $r$ . Find  $r^2$ .
74. Two circles of radius 5 are externally tangent to each other and are internally tangent to a circle of radius 13 at points A and B, as shown in the diagram. The distance AB can be written in the form  $\frac{m}{n}$ , where m and n are relatively prime positive integers. What is  $m + n$ ?



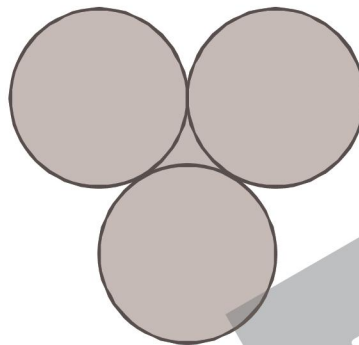
75. In the figure below,  $N$  congruent semicircles are drawn along a diameter of a large semicircle, with their diameters covering the diameter of the large semicircle with no overlap. Let  $A$  be the combined area of the small semicircles and  $B$  be the area of the region inside the large semicircle but outside the small semicircles. The ratio  $A : B$  is  $1 : 18$ . What is  $N$ ?



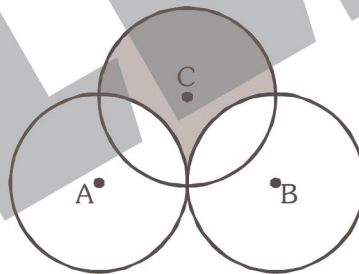
76. Seven cookies of radius 1 inch are cut from a circle of cookie dough, as shown. Neighboring cookies are tangent, and all except the center cookie are tangent to the edge of the dough. The leftover scrap is reshaped to form another cookie of the same thickness. If the radius in inches of the scrap cookie is  $R$ . Find  $4R^2$ .



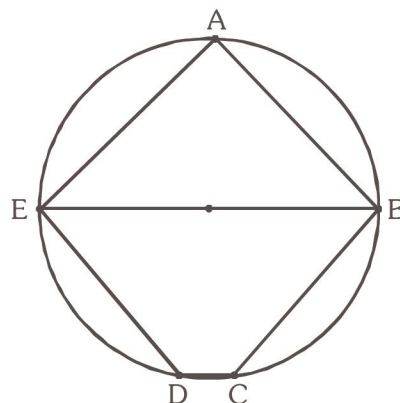
77. Six points are equally spaced around a circle of radius 1. Three of these points are the vertices of a triangle that is neither equilateral nor isosceles. If the area of this triangle is  $A$ , find  $\frac{4A}{\sqrt{3}}$ .
78. Mary divides a circle into 12 sectors. The central angles of these sectors, measured in degrees, are all integers and they form an arithmetic sequence. What is the degree measure of the smallest possible sector angle?
79. Externally tangent circles with centers at points  $A$  and  $B$  have radii of lengths 5 and 3, respectively. A line externally tangent to both circles intersects ray  $AB$  at point  $C$ . What is  $BC$ ?
80. Three circles with radius 2 are mutually tangent. If the total area of the circles and the region bounded by them, as shown in the figure is  $10\pi + 4\sqrt{k}$ , find  $k$ .



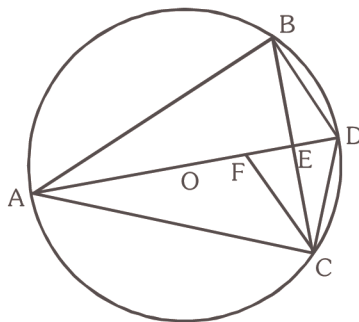
81. Circles  $A$ ,  $B$ , and  $C$  each have radius 1. Circles  $A$  and  $B$  share one point of tangency. Circle  $C$  has a point of tangency with the midpoint of  $\overline{AB}$ . What is the area inside Circle  $C$  but outside circle  $A$  and circle  $B$ ?



82. In the given circle, the diameter  $\overline{EB}$  is parallel to  $\overline{DC}$ , and  $\overline{AB}$  is parallel to  $\overline{ED}$ . The angles  $AEB$  and  $ABE$  are in the ratio 4 : 5. If the degree measure of angle  $BCD$  is  $k$ , find  $\frac{k}{5}$ .



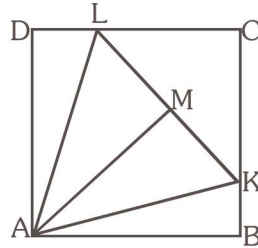
83. A circle with center  $O$  has area  $156\pi$ . Triangle  $ABC$  is equilateral,  $\overline{BC}$  is a chord on the circle,  $OA = 4\sqrt{3}$ , and point  $O$  is outside  $\triangle ABC$ . What is the side length of  $\triangle ABC$ ?
84. In the figure below,  $ABC$  is an isosceles triangle inscribed in a circle with centre  $O$  and diameter  $AD$ , with  $AB = AC$ .  $AD$  intersects  $BC$  at  $E$ , and  $F$  is the midpoint of  $OE$ . Given that  $BD$  is parallel to  $FC$  and  $BC = 2\sqrt{5}$  cm, find square of the length  $CD$  (in cm).



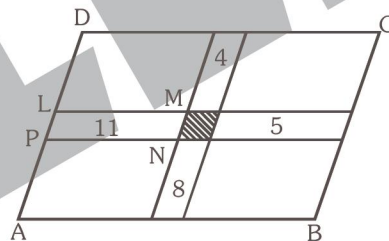
### 4.3 Polygons

#### EXERCISE - 4.3

1. ABCD is a square and K and L are points on BC and DC. AM is perpendicular to LK.  $\angle AKM = \angle AKB$ . Find the measure of  $\angle LAK$



- (1)  $45^\circ$  (2)  $70^\circ$  (3)  $55^\circ$  (4)  $35^\circ$
2. PQRS is a parallelogram. MP and NP divide  $\angle SPQ$  into three equal parts ( $\angle MPQ > \angle NPQ$ ) and MQ and NQ divide  $\angle RQP$  into 3 equal parts ( $\angle MQP > \angle NQP$ ). If  $k(\angle PNQ = \angle PMQ)$  then  $k =$
- (1)  $\frac{1}{2}$  (2) 1 (3)  $\frac{3}{2}$  (4)  $\frac{1}{3}$
3. ABCD is a trapezium with  $AB \parallel CD$  and  $AB < DC$ . AC and BD intersect at E,  $EF \parallel AB$ , intersecting BC at F. Given that  $AB = 20$ ,  $CD = 80$ ,  $BC = 100$ , then EF is
- (1) 10 (2) 12 (3) 16 (4) 18
4. In the adjoining figure ABCD is a parallelogram of perimeter 21. It is subdivided into smaller parallelograms by 7 drawing lines parallel to the sides. The numbers shown are the respective perimeters of the parallelograms in which they are marked. (For example the perimeter of the parallelogram LMNP is 11). Find the perimeter of the shaded parallelogram.



- (1) 7 (2) 8 (3) 10 (4) 11
5. A polygonal prism has 2010 edges. The number of the faces of the prism will be
- (1) 772 (2) 670 (3) 1005 (4) 672
6. The diagonals of a convex quadrilateral are perpendicular. If  $AB = 4$ ,  $AD = 5$ ,  $CD = 6$ , then length of BC is \_\_\_\_\_.
- (1)  $2\sqrt{3}$  (2)  $4\sqrt{3}$  (3)  $3\sqrt{3}$  (4)  $5\sqrt{3}$
7. The quadrilateral ABCD is inscribed in a circle. The diagonals AC and BD cut at Q. The sides DA and CB are produced to meet at P. If  $CD = CP = DQ$  then the measure of  $\angle CAD$  is
- (1)  $45^\circ$  (2)  $70^\circ$  (3)  $60^\circ$  (4)  $55^\circ$
8. ABCD is a convex quadrilateral in which  $AD = \sqrt{3}$ ,  $\angle A = 60^\circ$ ,  $\angle D = 120^\circ$  and  $AB + CD = 2AD$ . M is the midpoint of BC. Then DM =
- (1)  $\frac{\sqrt{3}}{2}$  (2)  $\frac{3}{2}$  (3)  $\frac{2}{3}$  (4)  $\frac{2}{\sqrt{3}}$

9. In a rectangle ABCD where  $AB = 6$ ,  $BC = 3$ , point P is chosen on AB such that  $\angle APD = 2\angle CPB$ . Then  $AP =$ \_\_\_\_\_.

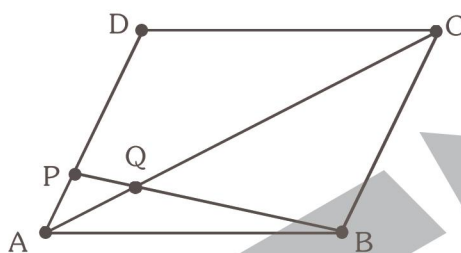
(1)  $(1 - \sqrt{7})$  (2)  $(4 - \sqrt{7})$  (3)  $(3 - \sqrt{7})$  (4)  $(5 - \sqrt{7})$

10. ABCD is a parallelogram P is a point on AD such that  $\frac{AP}{AD} = \frac{1}{2013}$ . Q is the point of intersection of AC and BP.

Calculate  $\frac{AQ}{AC} =$ \_\_\_\_\_.

11. ABCD is a square. With centres B and C and radius equal to the side of the square, circles are drawn to cut one another at E inside the square.  $\angle BDE$  is equal to

(1)  $22\frac{1}{2}^\circ$  (2)  $30^\circ$  (3)  $15^\circ$  (4)  $37^\circ$



(1)  $\frac{1}{2014}$  (2)  $\frac{2}{2014}$  (3)  $\frac{3}{2014}$  (4)  $\frac{4}{2014}$

12. ABCD is a rectangle in which  $AB = 8$ ,  $AD = 9$ . E is on AD such that  $DE = 4$ . H is on BC such that  $BH = 6$ . EC and AH cut at G. GF is drawn perpendicular to AD produced. Then  $GF =$

(1) 20 (2) 22 (3) 18 (4) 15

13. In the right triangle ABC,  $\angle C = 90^\circ$ ,  $BC = 12$  cm,  $AC = 6$  cm, the perpendicular bisector of AB intersects AB and BC at D and E respectively. Find CE.

(1) 4.5 (2) 3.5 (3) 2.5 (4) 1.5

14. In the rectangle ABCD,  $CE \perp DB$  at E,  $BE = \frac{1}{4}BD$  and  $CE = 5$  cm. Find the length of AC.

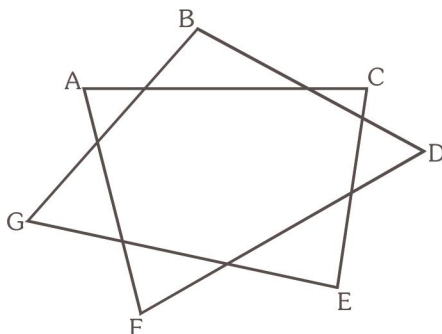
(1)  $\frac{20\sqrt{3}}{3}$  (2)  $\frac{25\sqrt{3}}{3}$  (3)  $\frac{26\sqrt{3}}{3}$  (4)  $\frac{29\sqrt{3}}{3}$

15. ABCD is a rectangle with  $AD = 2$ ;  $AB = 4$ . P is on AB such that  $AP : PB = 2 : 1$ ,  $CE \perp DP$  at E. Find CE.

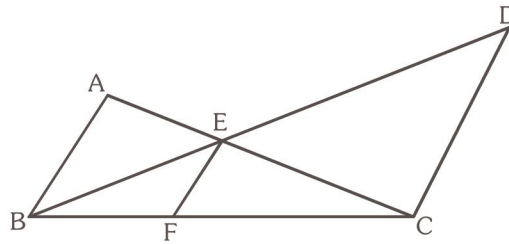
16. In the figure below, if

$$\angle A + \angle B + \angle C + \angle D + \angle E + \angle F + \angle G = 10x \text{ degrees,}$$

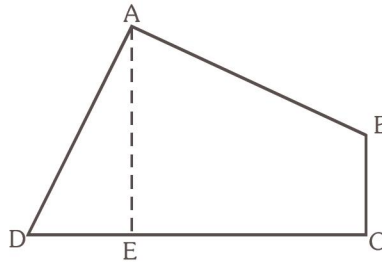
then what is x?



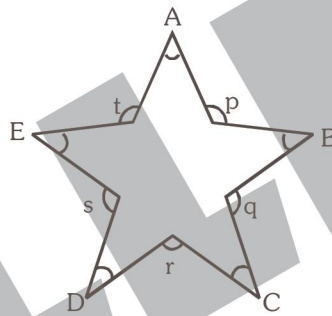
17. In the diagram,  $AB \parallel EF \parallel DC$ . Given that  $AC + BD = 250$ ,  $BC = 100$  and  $EC + ED = 150$  find  $CF$ .



18. In the following figure,  $AD = AB$ ,  $\angle DAB = \angle DCB = \angle AEC = 90^\circ$  and  $AE = 5$ . Find the area of the quadrangle ABCD.

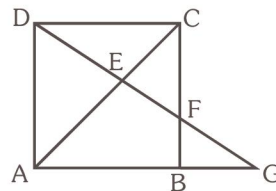


19. Consider the following "star" figure.

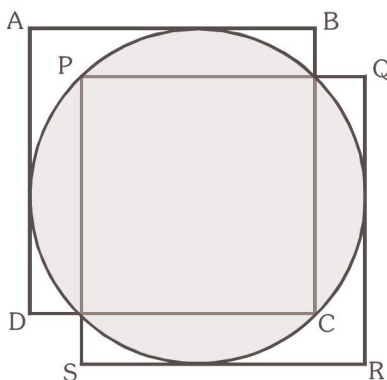


Given that  $\angle p + \angle q + \angle r + \angle s + \angle t = 500^\circ$  and  $\angle A + \angle B + \angle C + \angle D + \angle E = x^\circ$ , find the value of  $180 - x$ .

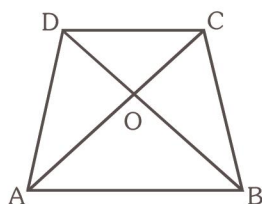
20. Let ABCDEF be a hexagon such that the diagonals AD, BE and CF intersect at the point O, and the area of the triangle formed by any three adjacent points is 2 (for example, area of  $\triangle BCD$  is 2). Find the area of the hexagon.
21. In the diagram below, ABCD is a square. The points A, B and G are collinear. The line segments AC and DG intersect at E and the line segments DG and BC intersect at F. Suppose that  $DE = 15$  cm,  $EF = 9$  cm, and  $FG = x$  cm. Find the value of  $x$ .



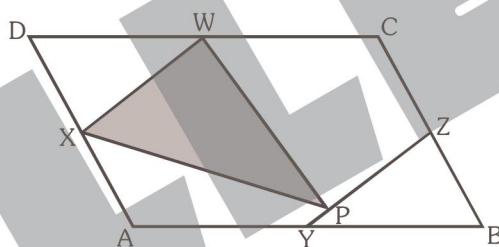
22. Let P be a 30-sided polygon inscribed in a circle. There are N number of triangles whose vertices are the vertices of P such that any two vertices of each triangle are separated by at least three other vertices of P. Then determine value of  $\frac{N}{100}$
23. The diagram shows two identical squares, ABCD and PQRS, overlapping each other in such a way that their edges are parallel, and a circle of radius  $(2 - \sqrt{2})$  cm covered within these squares. Find the length of the square ABCD.



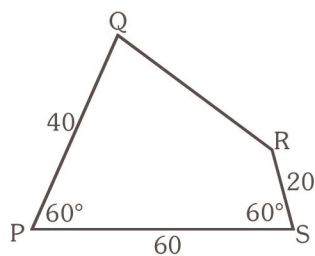
24. In the following figure,  $AB \parallel DC$ ,  $AB = b$ ,  $CD = a$  and  $a < b$ . Let  $S$  be the area of the trapezium  $ABCD$ . Suppose the area of  $\triangle BOC$  is  $\frac{2S}{9}$ . Find the value of  $b/a$ .



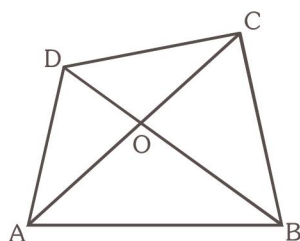
25.  $W, X, Y$  and  $Z$  are the midpoints of the four sides of parallelogram  $ABCD$ .  $P$  is a point on the line segment  $YZ$ . What percent of the area of parallelogram  $ABCD$  is triangle  $PXW$ ?



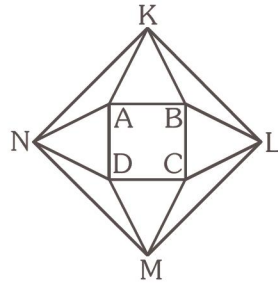
26. Four rods are connected together with flexible joints at their ends to make a quadrilateral as shown. Rods  $PQ = 40$  cm,  $RS = 20$  cm,  $PS = 60$  cm and  $\angle QPS = \angle RSP = 60^\circ$ . Find the value of  $x$  where  $x^\circ = 180^\circ - \angle QRS$



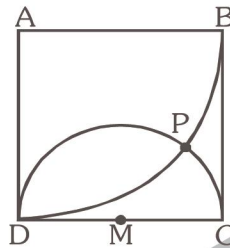
27. The diagram below shows a quadrilateral  $ABCD$  where  $AB = 10$ ,  $BC = 6$ ,  $CD = 8$  and  $DA = 2$ . The diagonals  $AC$  and  $BD$  intersect at the point  $O$  and that  $\angle COB = 45^\circ$ . Find the area of the quadrilateral  $ABCD$ .



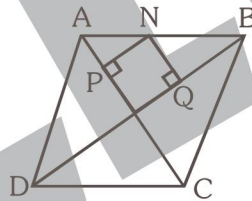
28. Points K, L, M, and N lie in the plane of the square ABCD so that AKB, BLC, CMD, and DNA are equilateral triangles. If ABCD has an area of 16, If the area of KLMN is  $a + b\sqrt{3}$  then find b.



29. Square ABCD has sides of length 4, and M is the midpoint of  $\overline{CD}$ . A circle with radius 2 and center M intersects a circle with radius 4 and center A at points P and D. If the distance from P to  $\overline{AD}$  is  $\frac{a}{b}$  then  $a + b = ?$



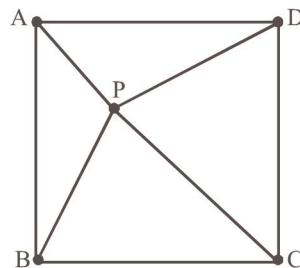
30. Let ABCD be a rhombus with  $AC = 16$  and  $BD = 30$ . Let N be a point on  $\overline{AB}$ , and let P and Q be the feet of the perpendiculars from N to  $\overline{AC}$  and  $\overline{BD}$ , respectively. Which of the following is closest to the minimum possible value of PQ?



31. In rectangle ABCD, points F and G lie on  $\overline{AB}$  so that  $AF = FG = GB$  and E is the midpoint of  $\overline{DC}$ . Also,  $\overline{AC}$  intersects  $\overline{EF}$  at H and  $\overline{EG}$  at J. The area of the rectangle ABCD is 70. Find the area of triangle EHJ.

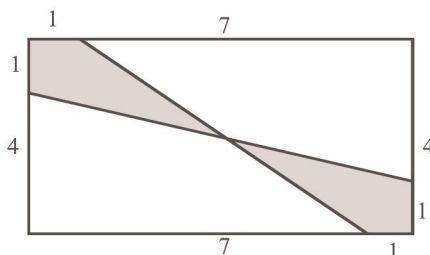
32. Square ABCD has side length 30. Point P lies inside the square so that  $AP = 12$  and  $BP = 26$ . The centroids of  $\triangle ABP$ ,  $\triangle BCP$ ,  $\triangle CDP$ , and  $\triangle DAP$  are the vertices of a convex quadrilateral. The area of that quadrilateral is A.

What is  $\frac{A}{4}$ ?

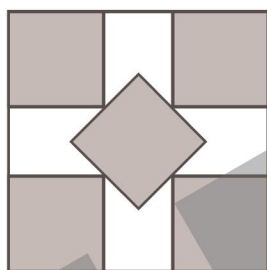


33. The ratio of the short side of a certain rectangle to the long side is equal to the ratio of the long side to the diagonal. The square of the ratio of the short side to the long side of this rectangle is in the form of  $\frac{\sqrt{a-b}}{c}$ . What is  $(a + b - c)^2$ ?

34. The area of the shaded region of the given  $8 \times 5$  rectangle is an improper fraction with co-prime digits. What is the sum of numerator and denominator?



35. The five small shaded squares inside this unit square are congruent and have disjoint interiors. The midpoint of each side of the middle square coincides with one of the vertices of the other four small squares as shown. The common side length is  $\frac{a-\sqrt{2}}{b}$ , where  $a$  and  $b$  are positive integers. What is  $a + b$ ?



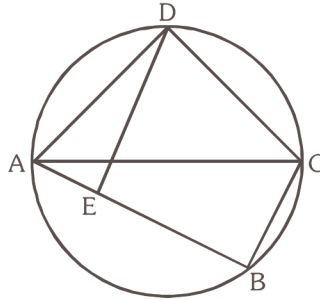
36. Let  $ABCD$  be a square. Let  $E, F, G$  and  $H$  be the centers, respectively, of equilateral triangles with bases  $\overline{AB}$ ,  $\overline{BC}$ ,  $\overline{CD}$  and  $\overline{DA}$ , each exterior to the square. The ratio of the area of square  $EFGH$  to the area of square  $ABCD$  is  $\frac{P+\sqrt{Q}}{R}$ . Find  $(P^3 + Q^2 + R)$ .
37. For some positive integers  $p$ , there is a quadrilateral  $ABCD$  with positive integer side lengths, perimeter  $p$ , right angles at  $B$  and  $C$ ,  $AB = 2$ , and  $CD = AD$ . How many different values of  $p < 2015$  possible?
38. In  $\triangle ABC$ ,  $\angle C = 90^\circ$  and  $AB = 12$ . Squares  $ABXY$  and  $ACWZ$  are constructed outside of the triangle. The points  $X, Y, Z$  and  $W$  lie on a circle. The perimeter of the triangle is in the form of  $a + b\sqrt{2}$ . What is  $(a + b)$ ?
39. Convex quadrilateral  $ABCD$  has  $AB = 9$  and  $CD = 12$ . Diagonals  $AC$  and  $BD$  intersect at  $E$ ,  $AC = 14$ , and  $\triangle AED$  and  $\triangle BEC$  have equal areas. What is  $7 \times AE$ ?
40. Trapezoid  $ABCD$  has  $AD \parallel BC$ ,  $BD = 1$ ,  $\angle DBA = 23^\circ$ , and  $\angle BDC = 46^\circ$ . The ratio  $BC : AD$  is  $9 : 5$ .  $CD = \frac{p}{q}$  where  $\gcd(p, q) = 1$ . What is  $p^2 + q^2$ ?
41. Five identical rectangles of area  $8 \text{ cm}^2$  are arranged into a large rectangle as shown.



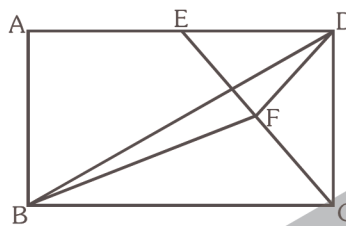
Find the perimeter of this large rectangle.

42. The lengths of the sides of a quadrilateral are 2006 cm, 2007 cm, 2008 cm and  $x$  cm. If  $x$  is an integer and its largest possible value is given as  $10^3a + b$  then find  $a + b$ .

43. Let ABCD be a quadrilateral inscribed in a circle with diameter AC, and let E be the foot of perpendicular from D onto AB, as shown in the figure below. If  $AD = DC$  and the area of quadrilateral ABCD is  $24 \text{ cm}^2$ , find the square of length of DE (in cm).



44. In the figure below, ABCD is a rectangle, E is the midpoint of AD and F is the midpoint of CE. If the area of triangle BDF is  $12 \text{ cm}^2$ , find the area of rectangle ABCD in  $\text{cm}^2$ .

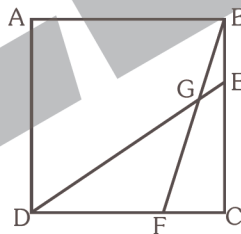


45. Given that ABCD is a square. Points E and F lie on the side BC and CD respectively, such that  $BE = CF = \frac{1}{3} AB$ .

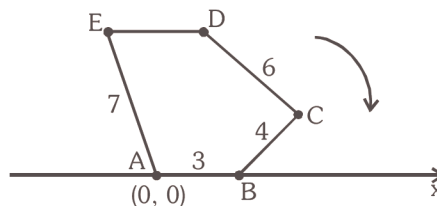
G is the intersection of BF and DE. If

$$\frac{\text{Area of ABGD}}{\text{Area of ABCD}} = \frac{m}{n}$$

is in its lowest term, find the value of  $m + n$ .

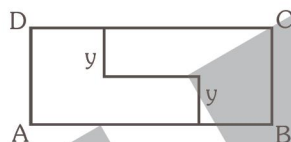


46. Rectangle ABCD has  $AB = 4$  and  $BC = 3$ . Segment EF is constructed through B so that EF is perpendicular to DB, and A and C lie on DE and DF, respectively. What is EF to the nearest integer?
47. As shown below, convex pentagon ABCDE has sides  $AB = 3$ ,  $BC = 4$ ,  $CD = 6$ ,  $DE = 3$ , and  $EA = 7$ . The pentagon is originally positioned in the plane with vertex A at the origin and vertex B on the positive x-axis. The pentagon is then rolled clockwise to the right along the x-axis. What is the length side which will touch the point  $x = 2009$  on the x-axis?

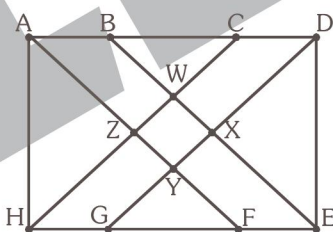


48. Rectangle ABCD has  $AB = 8$  and  $BC = 6$ . Point M is the midpoint of diagonal  $\overline{AC}$ , and E is on  $\overline{AB}$  with  $\overline{ME} \perp \overline{AC}$ . What is the area of  $\triangle AME$  to the nearest integer?

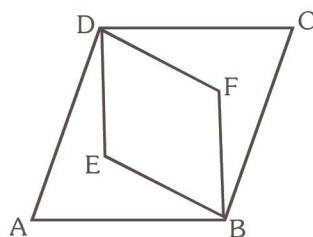
49. Each of the sides of a square  $S_1$  with area 16 is bisected, and a smaller square  $S_2$  is constructed using the bisection points as vertices. The same process is carried out on  $S_2$  to construct an even smaller square  $S_3$ . What is the area of  $S_3$ ?
50. Rectangle PQRS lies in a plane with  $PQ = RS = 2$  and  $QR = SP = 6$ . The rectangle is rotated  $90^\circ$  clockwise about R, then rotated  $90^\circ$  clockwise about the point S that moved to after the first rotation. What is the length of the path traveled by point P to the nearest integer? (Take  $\pi = 3.14$ )
51. Trapezoid ABCD has bases  $\overline{AB}$  and  $\overline{CD}$  and diagonals intersecting at K. Suppose that  $AB = 9$ ,  $DC = 12$ , and the area of  $\triangle AKD$  is 24. What is the area of trapezoid ABCD?
52. Quadrilateral ABCD has  $AB = BC = CD$ ,  $\angle ABC = 70^\circ$  and  $\angle BCD = 170^\circ$ . What is the degree measure of  $\angle BAD$ ?
53. The angles of quadrilateral ABCD satisfy  $\angle A = 2\angle B = 3\angle C = 4\angle D$ . What is the degree measure of  $\angle A/2$ , rounded to the nearest whole number?
54. The  $8 \times 18$  rectangle ABCD is cut into two congruent hexagons, as shown, in such a way that the two hexagons can be repositioned without overlap to form a square. What is  $y$ ?



- 55.** In rectangle ADEH, points B and C trisect  $\overline{AD}$ , and points G and F trisect  $\overline{HE}$ . In addition,  $AH = AC = 2$ . What is twice of the area of quadrilateral WXYZ shown in the figure?

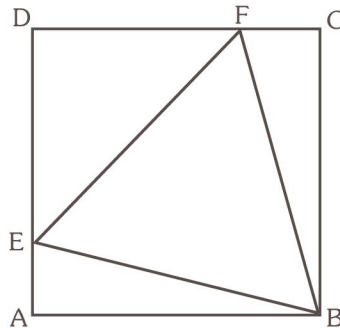


- 56.** What is the tens digit in the sum  $7! + 8! + 9! + \cdots + 2006!$ ?
- 57.** Rhombus ABCD is similar to rhombus BFDE. The area of rhombus ABCD is 24, and  $\angle BAD = 60^\circ$ . What is the area of rhombus BFDE?

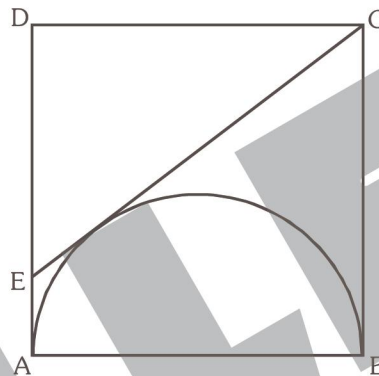


- 58.** In  $\triangle ABC$  we have  $AB = 25$ ,  $BC = 39$ , and  $AC = 42$ . Points  $D$  and  $E$  are on  $AB$  and  $AC$  respectively, with  $AD = 19$  and  $AE = 14$ . What is the ratio of the area of triangle  $ADE$  to the area of quadrilateral  $BCED$  is  $\frac{m}{n}$  then find  $m + n$

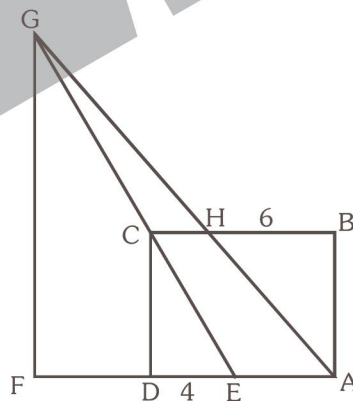
59. In trapezoid ABCD we have  $\overline{AB}$  parallel to  $\overline{DC}$ , E as the midpoint of  $\overline{BC}$ , and F as the midpoint of  $\overline{DA}$ . The area of ABEF is twice the area of FECD. What is  $AB/DC$ ?
60. Points E and F are located on square ABCD so that  $\triangle BEF$  is equilateral. What is the ratio of the area of  $\triangle DEF$  to that of  $\triangle ABE$ ?



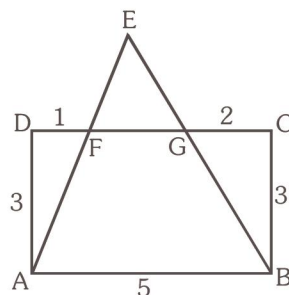
61. Square ABCD has side length 2. A semicircle with diameter  $\overline{AB}$  is constructed inside the square, and the tangent to the semicircle from C intersects side  $\overline{AD}$  at E. Find  $2\overline{CE}$ .



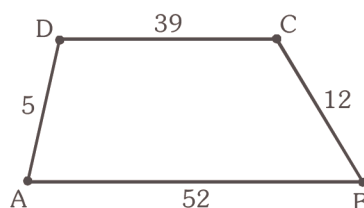
62. In rectangle ABCD, we have  $AB = 8$ ,  $BC = 9$ , H is on  $\overline{BC}$  with  $BH = 6$ , E is on  $\overline{AD}$  with  $DE = 4$ , line EC intersects line AH at G, and F is on line AD with  $\overline{GF} \perp \overline{AF}$ . Find the length GF.



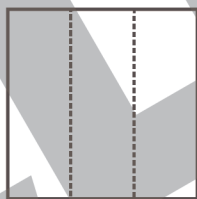
63. In rectangle ABCD,  $AB = 5$  and  $BC = 3$ . Points F and G are on  $\overline{CD}$  so that  $DF = 1$  and  $GC = 2$ . Lines AF and BG intersect at E. Then the area of  $\triangle AEB$  is given by S. Find the value of  $2S$ .



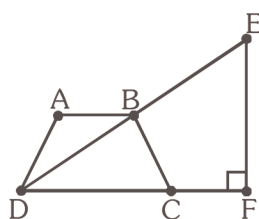
64. In trapezoid ABCD with bases AB and CD, we have  $AB = 52$ ,  $BC = 12$ ,  $CD = 39$ , and  $DA = 5$ . The area of ABCD is given by  $10A$ . Find the value of  $A$ .



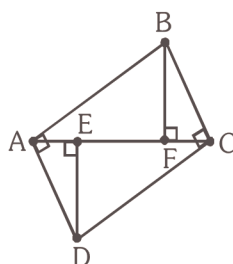
65. A regular octagon ABCDEFGH has sides of length two. Then the area of  $\triangle ADG$  is given by  $(a + b\sqrt{c})$ . Find the product of  $a$ ,  $b$  &  $c$ .
66. A regular octagon is formed by cutting an isosceles right triangle from each of the corners of a square with sides of length 20. Let the length of each side of the octagon is given by  $a(\sqrt{b} - c)$ . Find the product of  $a$ ,  $b$  and  $c$ .
67. A right circular cylinder with its diameter equal to its height is inscribed in a right circular cone. The cone has diameter 10 and altitude 12, and the axes of the cylinder and cone coincide. Then the radius of the cylinder is  $R$ . Find the value of  $(11R)$ .
68. In trapezoid ABCD,  $\overline{AB}$  and  $\overline{CD}$  are perpendicular to  $\overline{AD}$ , with  $AB + CD = BC$ ,  $AB < CD$ , and  $AD = 7$ . What is  $(4AB \cdot CD)$ ?
69. A square is cut into three rectangles along two lines parallel to a side, as shown. If the perimeter of each of the three rectangles is 24, then the area of the original square is



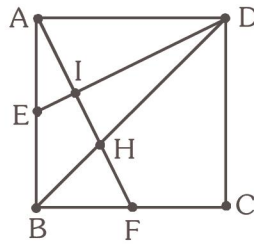
70. In the figure, ABCD is an isosceles trapezoid with side lengths  $AD = BC = 5$ ,  $AB = 4$ , and  $DC = 10$ . The point C is on  $\overline{DF}$  and B is the midpoint of hypotenuse  $\overline{DE}$  in the right triangle DEF. Then  $CF =$



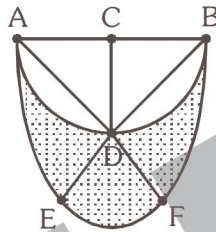
71. In the figure, ABCD is a quadrilateral with right angles at A and C. Points E and F are on  $\overline{AC}$ , and  $\overline{DE}$  and  $\overline{BF}$  are perpendicular to  $\overline{AC}$ . If  $AE = 3$ ,  $DE = 5$  and  $CE = 7$ , then the value of  $BF - \frac{1}{5}$  is



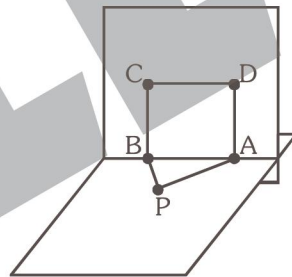
72. If ABCD is a  $2 \times 2$  square, E is the midpoint of  $\overline{AB}$ , F is the midpoint of  $\overline{BC}$ ,  $\overline{AF}$  and  $\overline{DE}$  intersect at I, and BD and AF intersect at H, and the area of quadrilateral BEIH is  $\frac{k}{2k+1}$ , find k.



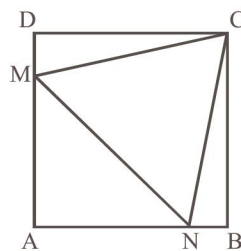
73. Semicircle  $\widehat{AB}$  has center C and radius 1. Point D is on  $\widehat{AB}$  and  $\overline{CD} \perp \overline{AB}$ . Extend  $\widehat{BD}$  and  $\widehat{AD}$  to E and F respectively, so that circular arcs  $\widehat{AE}$  and  $\widehat{BF}$  have B and A as their respective centers, circular arc EF has center D. The area of the shaded "smile", AEFBDA is  $(2\pi - \pi\sqrt{2} - k)$ , find k.



74. Triangle PAB and square ABCD are in perpendicular planes. Given that  $PA = 3$ ,  $PB = 4$ , and  $AB = 5$ , what is  $PD^2$ ?

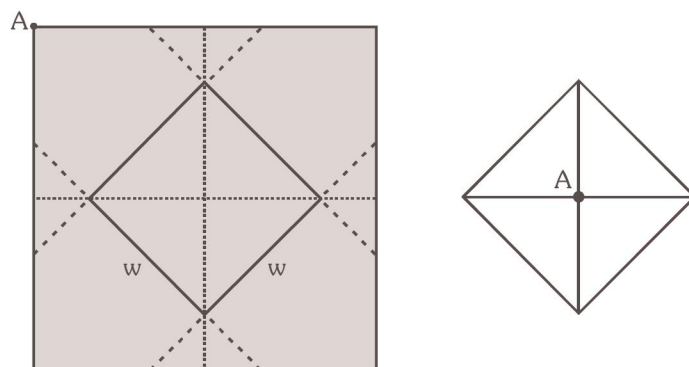


75. In quadrilateral ABCD, it is given that  $\angle A = 120^\circ$ , angles B and D are right angles,  $AB = 13$ , and  $AD = 46$ . Then  $AC =$
76. In the adjoining figure ABCD is a square and CMN is an equilateral triangle. If the area of ABCD is one square inch, and the area of CMN in square inches is given by  $(2\sqrt{a} - b)$ . Find the product of a and b.

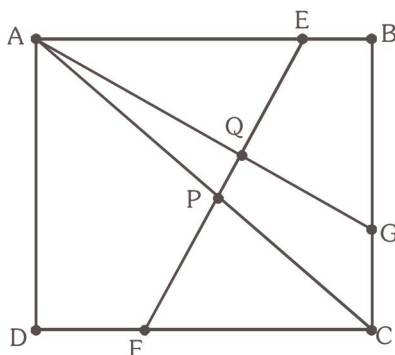


77. A three-dimensional rectangular box with dimensions X, Y and Z has faces whose surface areas are 24, 24, 48, 48, 72, and 72 square units. What is  $X + Y + Z$ ?

78. A closed box with a square base is to be wrapped with a square sheet of wrapping paper. The box is centered on the wrapping paper with the vertices of the base lying on the midlines of the square sheet of paper, as shown in the figure on the left. The four corners of the wrapping paper are to be folded up over the sides and brought together to meet at the center of the top of the box, point A in the figure on the right. The box has base length 2 meter and height 5 meter. What is the area of the sheet of wrapping paper?

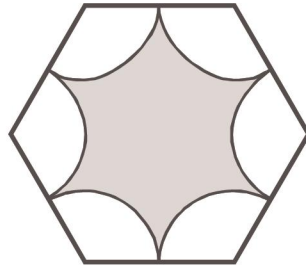


79. Let  $ABCDEF$  be a regular hexagon with side length 1. Denote by  $X$ ,  $Y$  and  $Z$  the midpoints of  $\overline{AB}$ ,  $\overline{CD}$ , and  $\overline{EF}$ , respectively. If the area of the convex hexagon whose interior is the intersection of the interiors of  $\triangle ACE$  and  $\triangle XYZ$  is given by  $\frac{a\sqrt{b}}{32}$ . Find the product of  $a$  and  $b$ .
80. In rectangle  $ABCD$ ,  $AB = 6$  and  $BC = 3$ . Point  $E$  between  $B$  and  $C$ , and point  $F$  between  $E$  and  $C$  are such that  $BE = EF = FC$ . Segments  $\overline{AE}$  and  $\overline{AF}$  intersect  $\overline{BD}$  at  $P$  and  $Q$ , respectively. The ratio  $BP : PQ : QD$  can be written as  $r : s : t$ , where the greatest common factor of  $r$ ,  $s$  and  $t$  is 1. What is  $r + s + t$ ?
81. A quadrilateral is inscribed in a circle of radius  $20\sqrt{2}$ . Three of the sides of this quadrilateral have length 20. What is the length of the fourth side?
82. Chetan decided to fence in his rectangular garden. He bought 20 fence posts, placed one on each of the four corners, and spaced out the rest evenly along the edges of the garden, leaving exactly 4 yards between neighboring posts. The longer side of his garden, including the corners, has twice as many posts as the shorter side, including the corners. If the area, in square yards, of Carls garden is given by  $A$ . Find  $[400 - A]$ .
83. Rectangle  $ABCD$  has  $AB = 5$  and  $BC = 4$ . Point  $E$  lies on  $\overline{AB}$  so that  $EB = 1$ , point  $G$  lies on  $\overline{BC}$  so that  $CG = 1$ , and point  $F$  lies on  $\overline{CD}$  so that  $DF = 2$ . Segments  $\overline{AG}$  and  $\overline{AC}$  intersect  $\overline{EF}$  at  $Q$  and  $P$ , respectively. What is the value of  $91\left(1 - \frac{PQ}{EF}\right)$ ?

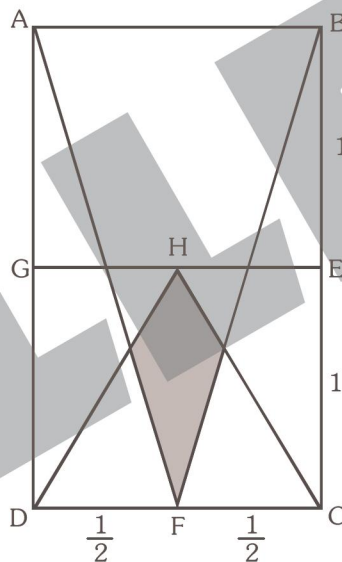


84. The ratio of the length to the width of a rectangle is  $4 : 3$ . If the rectangle has diagonal of length  $d$ , then the area may be expressed as  $kd^2$  for some constant  $k$ . Find the value of  $100k$ ?

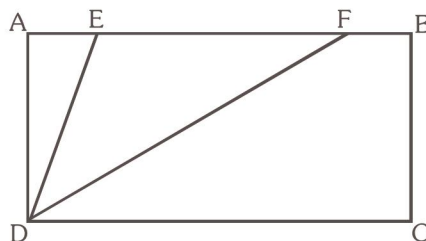
85. For some positive integers  $p$ , there is a quadrilateral  $ABCD$  with positive integer side lengths, perimeter  $p$ , right angles at  $B$  and  $C$ ,  $AB = 2$ , and  $CD = AD$ . How many different values of  $p < 2015$  are possible?
86. A rectangular box measures  $a \times b \times c$ , where  $a$ ,  $b$ , and  $c$  are integers and  $1 \leq a \leq b \leq c$ . The volume and surface area of the box are numerically equal. How many ordered triples  $(a, b, c)$  are possible?
87. A regular hexagon has side length 6. Congruent arcs with radius 3 are drawn with the center at each of the vertices, creating circular sectors as shown. The region inside the hexagon but outside the sectors is shaded as shown. The area of the shaded region can be written as  $(a\sqrt{3} - \pi b)$ . Then find the sum of  $a$  and  $b$ .



88. In rectangle  $ABCD$ ,  $AB = 1$ ,  $BC = 2$ , and points  $E$ ,  $F$ , and  $G$  are midpoints of  $\overline{BC}$ ,  $\overline{CD}$ , and  $\overline{AD}$ , respectively. Point  $H$  is the midpoint of  $\overline{GE}$ . What is the reciprocal of area of the shaded region?

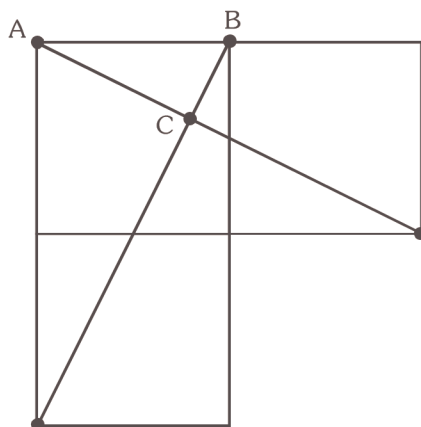


89. In rectangle  $ABCD$ ,  $DC = 2CB$  and points  $E$  and  $F$  lie on  $\overline{AB}$  so that  $\overline{ED}$  and  $\overline{FD}$  trisect  $\angle ADC$  as shown. If the ratio of the area of  $\triangle DEF$  to the area of rectangle  $ABCD$  is given by  $A$ . Find the value of  $10\sqrt{3}A$ .

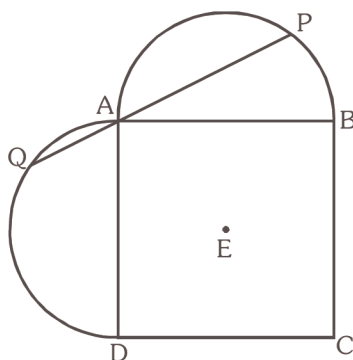


90. Trapezoid  $ABCD$  has parallel sides  $\overline{AB}$  of length 33 and  $\overline{CD}$  of length 21. The other two sides are of lengths 10 and 14. The angles at  $A$  and  $B$  are acute. What is the length of the shorter diagonal of  $ABCD$ ?
91. All diagonals are drawn in a regular octagon. At how many distinct points in the interior of the octagon (not on the boundary) do two or more diagonals intersect?

92. A wire is cut into two pieces, one of length  $a$  and the other of length  $b$ . The piece of length  $a$  is bent to form an equilateral triangle, and the piece of length  $b$  is bent to form a regular hexagon. The triangle and the hexagon have equal area. What is  $\frac{4a^2}{b^2}$ ?
93. Three unit squares and two line segments connecting two pairs of vertices are shown. If the area of  $\triangle ABC$  is  $N$ , find  $5N$ .

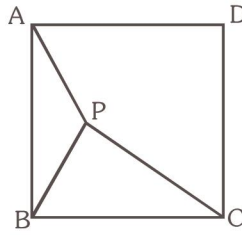


94. In rectangle  $ABCD$ ,  $AB = 6$ ,  $AD = 30$ , and  $G$  is the midpoint of  $\overline{AD}$ . Segment  $AB$  is extended 2 units beyond  $B$  to point  $E$ , and  $F$  is the intersection of  $\overline{ED}$  and  $\overline{BC}$ . If the area of  $BFDG$  is  $X$ , find  $[X]$ .
95. Square  $EFGH$  has one vertex on each side of square  $ABCD$ . Point  $E$  is on  $\overline{AB}$  with  $AE = 7$ .  $EB$ . If the ratio of the area of  $EFGH$  to the area of  $ABCD$  is  $\frac{m}{n}$ , where  $m, n$  are co-prime, find  $m + n$ .
96. Rectangle  $ABCD$  has  $AB = 6$  and  $BC = 3$ . Point  $M$  is chosen on side  $AB$  so that  $\angle AMD = \angle CMD$ . What is the degree measure of  $\angle AMD$ ?
97. Points  $A$  and  $C$  lie on a circle centered at  $O$ , each of  $\overline{BA}$  and  $\overline{BC}$  are tangent to the circle, and  $\triangle ABC$  is equilateral. The circle intersects  $\overline{BO}$  at  $D$ . What is  $BD + BO$ ?
98. Suresh has 30 thin rods, one each of every integer length from 1 cm through 30 cm. She places the rods with lengths 3 cm, 7 cm, and 15 cm on a table. She then wants to choose a fourth rod that she can put with these three to form a quadrilateral with positive area. How many of the remaining rods can she choose as the fourth rod?
99. Equiangular hexagon  $ABCDEF$  has side lengths  $AB = CD = EF = 1$  and  $BC = DE = FA = r$ . The area of  $\triangle ACE$  is 70% of the area of the hexagon. What is the sum of all possible values of  $r$ ?
100. In the following diagram,  $ABCD$  is a square, and  $E$  is the center of the square  $ABCD$ .  $P$  is a point on a semi-circle with diameter  $AB$ .  $Q$  is a point on a semi-circle with diameter  $AD$ . Moreover,  $Q, A$  and  $P$  are collinear (that is, they are on the same line). Suppose  $QA = 14$  cm,  $AP = 46$  cm, and  $AE = x$  cm. Find the value of  $x$ .



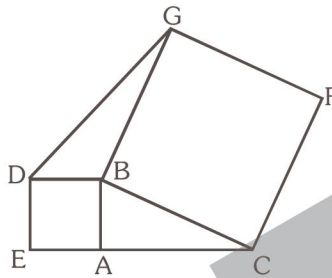
- 101.** In the following diagram, ABCD is a square with  $PA = a$ ,  $PB = 2a$  and  $PC = 3a$ .

Find the value of  $x$  where  $x^\circ = 180^\circ - \angle APB$ .



- 102.** A solid cube of side length 1 is removed from each corner of a solid cube of side length 3. How many edges does the remaining solid have?

- 103.** ABC is a right-angled triangle with  $\angle BAC = 90^\circ$ . A square is constructed on the side AB and BC as shown. The area of the square ABDE is  $8 \text{ cm}^2$  and the area of the square BCFG is  $26 \text{ cm}^2$ . Find the area of triangle DBG in  $\text{cm}^2$ .



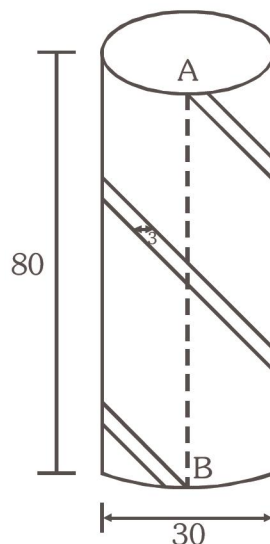
- 104.** One dimension of a cube is increased by 1, another is decreased by 1, and the third is left unchanged. The volume of the new rectangular solid is 5 less than that of the cube. If the volume of the cube is  $V$ . Find  $\sqrt[3]{V}$

- 105.** A cylindrical tank with radius 4 feet and height 9 feet is lying on its side. The tank is filled with water to a depth of 2 feet. If the volume of the water, in cubic feet is  $x\pi + y\sqrt{3}$  then find  $x + y$ ?

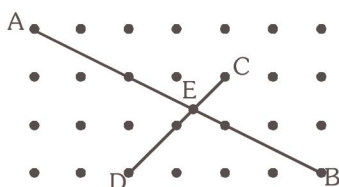
- 106.** A sphere is inscribed in a cube that has a surface area of 24 square meters. A second cube is then inscribed within the sphere. What is the surface area in square meters of the inner cube?

- 107.** Centers of adjacent faces of a unit cube are joined to form a regular octahedron. If the volume of this octahedron is  $V$ , then find the value of  $6V$ ?

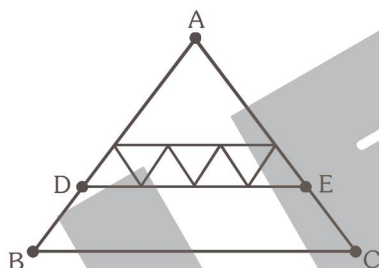
- 108.** A white cylindrical silo has a diameter of 30 feet and a height of 80 feet. A red stripe with a horizontal width of 3 feet is painted on the silo, as shown, making two complete revolutions around it. Find  $\frac{1}{3}$ rd of the area of the stripe in square feet.



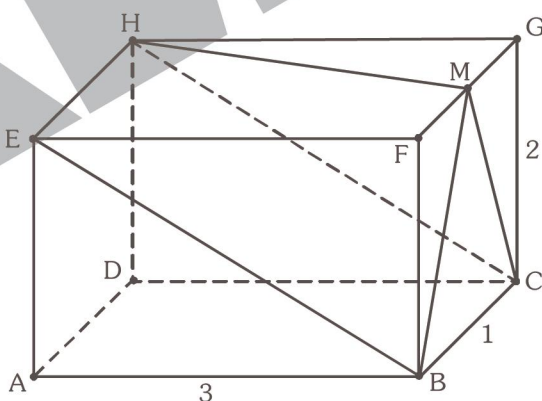
- 109.** An ice cream cone consists of a sphere of vanilla ice cream and a right circular cone that has the same diameter as the sphere. If the ice cream melts, it will exactly fill the cone. Assume that the melted ice cream occupies 75% of the volume of the frozen ice cream. If the ratio of the cones height to its radius is given by  $a/b$ . Find sum of  $a$  &  $b$ .
- 110.** The diagram show 28 lattice points, each one unit from its nearest neighbors. Segment AB meets segment CD at E. Then the length of segment AE is given by L. Find the value of  $\left(\frac{9L^2}{5}\right)$ .



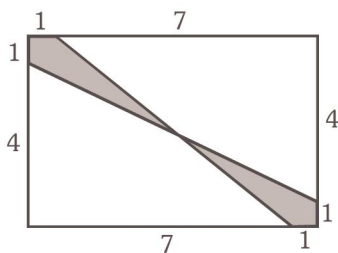
- 111.** All of the triangles in the diagram below are similar to isosceles triangle ABC, in which  $AB = AC$ . Each of the 7 smallest triangles has area 1, and  $\triangle ABC$  has area 40. What is the area of trapezoid DBCE?



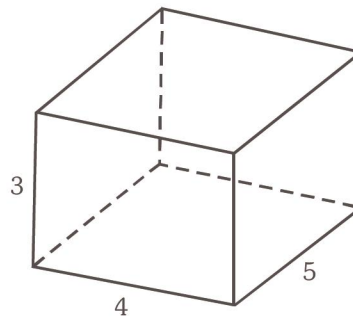
- 112.** In the rectangular parallelepiped shown,  $AB = 3$ ,  $BC = 1$ , and  $CG = 2$ . Point M is the midpoint of  $\overline{FG}$ . What is the volume of the rectangular pyramid with base BCHE and apex M?



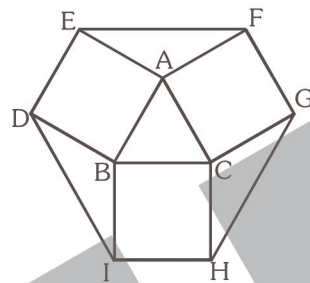
- 113.** If A is the area of the shaded region of the given  $8 \times 5$  rectangle. Find the value of  $4A$ .



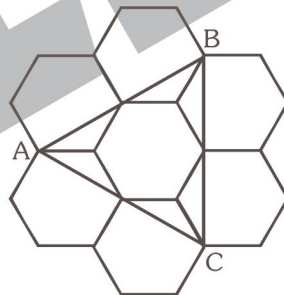
- 114.** When the centers of the faces of the right rectangular prism shown below are joined to create an octahedron, What is the volume of the octahedron?



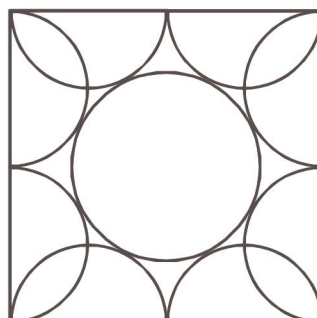
- 115.** Equilateral  $\triangle ABC$  has side length 1, and squares  $ABDE$ ,  $BCHI$ ,  $CAFG$  lie outside the triangle. If the area of hexagon  $DEFGHI$  is  $A$ . Then find the value of  $(A - \sqrt{3})^3$ .



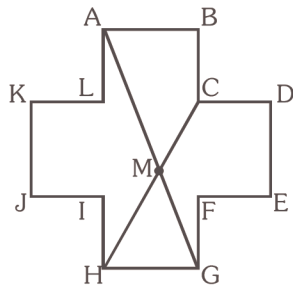
- 116.** Six regular hexagons surround a regular hexagon of side length 1 as shown. What is the square of area of  $\triangle ABC$ ?



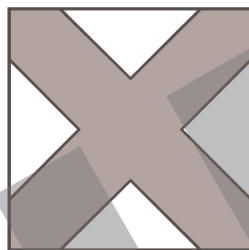
- 117.** Eight semicircles line the inside of a square with side length 2 as shown. If the radius of the circle tangent to all of these semicircles is given by  $\frac{\sqrt{a}-1}{b}$ , where  $a$  and  $b$  are prime number. Find the product of  $a$  and  $b$ .



- 118.** Convex quadrilateral ABCD has  $AB = 9$  and  $CD = 12$ . Diagonals AC and BD intersect at E,  $AC = 14$ , and  $\triangle AED$  and  $\triangle BEC$  have equal areas. What is AE?
- 119.** Consider the 12-sided polygon ABCDEFGHIJKL, as shown. Each of its sides has length 4, and each two consecutive sides form a right angle. Suppose that  $\overline{AG}$  and  $\overline{CH}$  meet at M. What is the area of quadrilateral ABCM to the nearest integer?



- 120.** A paint brush is swept along both diagonals of a square to produce the symmetric painted area, as shown. Half the area of the square is painted. If the ratio of the side length of the square to the brush width is  $m\sqrt{2} + n$  then find the value of  $(m^2n)^2$ ?



ANSWER KEY

EXERCISE - 4.1

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| Ans. | 3 | 4 | 2 | 1 | 2 | 4 | 3 | 1 | 2 | 1  | 1  | 3  | 1  | 1  |

|           |        |        |        |                                                             |        |        |
|-----------|--------|--------|--------|-------------------------------------------------------------|--------|--------|
| 15.8      | 16.0   | 17.70  | 19.30  | 20. $AC = 3\sqrt{5}$ , $AB = \sqrt{13}$ , $BC = 2\sqrt{13}$ |        |        |
| 21. 4 : 1 | 26.20  | 27.48  | 28.54  | 29.33                                                       | 30.15  | 31.01  |
| 32.12     | 33.87  | 34.03  | 35.12  | 36.26                                                       | 37.30  | 38.20  |
| 39.01     | 40.03  | 41.06  | 42.35  | 43.20                                                       | 44.02  | 45.50  |
| 46.15     | 47.02  | 48.70  | 49.77  | 50.52                                                       | 51.12  | 52.03  |
| 53.20     | 54.18  | 55.14  | 56.02  | 57.75                                                       | 58.12  | 59.80  |
| 60.37     | 61.05  | 62.48  | 63.05  | 64.14                                                       | 65.56  | 66.34  |
| 67.36     | 68.21  | 69.15  | 70.30  | 71.10                                                       | 72.48  | 73.00  |
| 74.08     | 75.12  | 76.04  | 77.27  | 78.02                                                       | 79.33  | 80.05  |
| 81.01     | 82.24  | 83.21  | 84.27  | 85.90                                                       | 86.02  | 87.03  |
| 88.82     | 89.03  | 90.15  | 91.01  | 92.12                                                       | 93.56  | 94.24  |
| 95.48     | 96.02  | 97.43  | 98.18  | 99.03                                                       | 100.23 | 101.04 |
| 102.65    | 103.27 | 104.12 | 105.05 | 106.09                                                      | 107.26 | 108.30 |
| 109.64    | 110.07 | 111.10 | 112.05 | 113.03                                                      | 114.36 | 115.40 |
| 116.15    | 117.12 | 118.10 | 119.60 | 120.15                                                      | 121.13 | 122.20 |
| 123.75    | 124.50 | 125.72 | 126.54 | 127.37                                                      | 128.33 | 129.75 |

EXERCISE - 4.2

| Que. | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Ans. | 2  | 2  | 1  | 2  | 1  | 1  | 4  | 1  | 3  | 4  | 4  | 2  | 1  | 4  | 3  |
| Que. | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |    |    |    |    |    |    |
| Ans. | 1  | 2  | 3  | 1  | 1  | 2  | 2  | 1  | 2  |    |    |    |    |    |    |

|       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|
| 25.20 | 26.57 | 27.36 | 28.60 | 29.04 | 30.05 | 31.07 |
| 32.49 | 33.17 | 34.03 | 35.53 | 36.72 | 37.66 | 38.40 |
| 39.36 | 40.61 | 41.30 | 42.63 | 43.08 | 44.16 | 45.54 |
| 46.65 | 47.60 | 48.78 | 49.45 | 50.8  | 51.35 | 52.82 |
| 53.10 | 54.11 | 55.04 | 56.02 | 57.23 | 58.44 | 59.24 |
| 60.02 | 61.60 | 62.04 | 63.05 | 64.03 | 65.06 | 66.92 |
| 67.01 | 68.73 | 69.02 | 70.06 | 71.27 | 72.16 | 73.03 |
| 74.69 | 75.19 | 76.08 | 77.02 | 78.08 | 79.12 | 80.03 |
| 81.02 | 82.26 | 83.06 | 84.06 |       |       |       |

## EXERCISE - 4.3

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| Ans. | 1 | 1 | 3 | 1 | 4 | 3 | 3 | 2 | 2 | 1  | 2  | 1  | 1  | 1  |

|                    |        |        |        |        |        |        |
|--------------------|--------|--------|--------|--------|--------|--------|
| 15. $\frac{12}{5}$ | 16.54  | 17.60  | 18.25  | 19.40  | 20.12  |        |
| 21.16              | 22.19  | 23.01  | 24.02  | 25.25  | 26.30  | 27.31  |
| 28.16              | 29.21  | 30.07  | 31.03  | 32.50  | 33.16  | 34.15  |
| 35.11              | 36.20  | 37.31  | 38.24  | 39.42  | 40.41  | 41.28  |
| 42.26              | 43.24  | 44.96  | 45.23  | 46.10  | 47.03  | 48.09  |
| 49.04              | 50.19  | 51.98  | 52.85  | 53.86  | 54.06  | 55.01  |
| 56.04              | 57.08  | 58.75  | 59.05  | 60.02  | 61.05  | 62.20  |
| 63.25              | 64.21  | 65.24  | 66.40  | 67.30  | 68.49  | 69.81  |
| 70.04              | 71.04  | 72.07  | 73.01  | 74.34  | 75.62  | 76.09  |
| 77.22              | 78.98  | 79.45  | 80.20  | 81.50  | 82.64  | 83.81  |
| 84.48              | 85.31  | 86.10  | 87.72  | 88.06  | 89.05  | 90.25  |
| 91.49              | 92.6   | 93.01  | 94.67  | 95.57  | 96.75  | 97.03  |
| 98.17              | 99.6   | 100.34 | 101.45 | 102.84 | 103.06 | 104.05 |
| 105.12             | 106.08 | 107.01 | 108.80 | 109.04 | 110.25 | 111.24 |
| 112.02             | 113.26 | 114.10 | 115.27 | 116.27 | 117.10 | 118.06 |
| 119.18             | 120.64 |        |        |        |        |        |